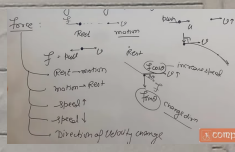
Newtons Laws of Motion



Force: Any kind of pull or push

→ It has tendency to change state of rest or motion.

→ It can bring any particle in motion from rest or vice versa.

→ It can change the speed or change direction of velocity.

→ It can deform any body.

→ Unit of force = kgm/sec^2 = Newton (N)

In CGS; $1N = \underbrace{1\frac{kgm}{sec^2}}_{\text{SI unit}} = \frac{10^3 gm\ 10^2 cm}{sec^2} = 10^5\left(\frac{gm\ cm}{sec^2}\right) = \underbrace{10^5 dyne}_{\text{CGS unit}}$

→ Dimension = $[MLT^{-2}]$

Types of forces:

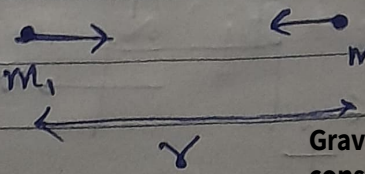
1. Gravitational force
2. Electromagnetic force
3. Strong Nuclear force
4. Weak Nuclear force

} Nuclear force ⇒ Acts in nucleus ⇒ Short Range

(I) Gravitational force:

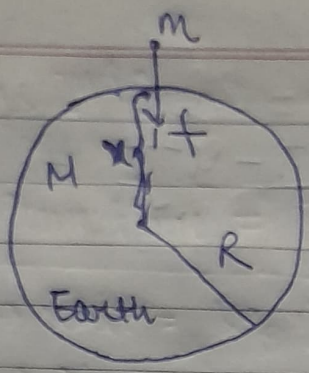
→ It is by virtue of masses

→ It is always attractive



$$F = \frac{G m_1 m_2}{r^2}$$

Gravitational constant $G = 6.67 \times 10^{-11} \frac{Nm^2}{kg^2} = \frac{20}{3} \times 10^{-11} \frac{Nm^2}{kg^2}$



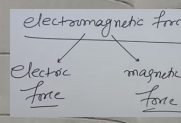
$R >>>> x$

$$F = \frac{GMm}{(R+x)^2} = \frac{GMm}{R^2} = mg$$

Here; $\frac{GM}{R^2}$ = const = $g = 9.8 m/s^2$
$= 32 ft/sec^2$
$\approx \pi^2 m/s^2$

mg = Weight

(II) Electromagnetic forces: It is by virtue of charge



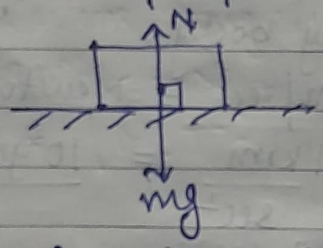
→ It can be attractive or repulsive

→ All contact forces are electromagnetic in nature

→ Two types: ① Electric force ② Magnetic Force.

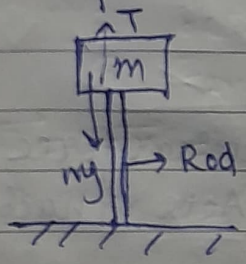
→ Examples:

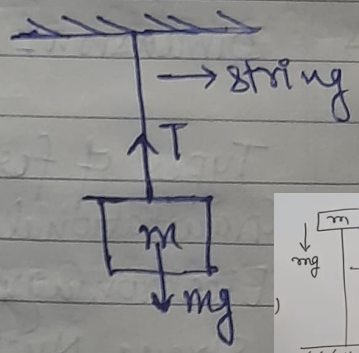
(a) Normal force: A perpendicular pushing force.

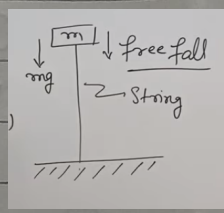


(b) Tension: Pulling force (Always)
(in case of String)

⇒ Rod can pull as well as push



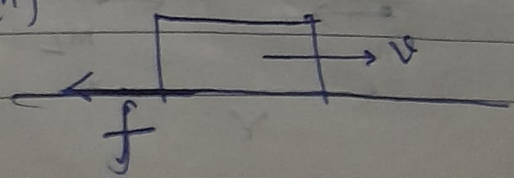




→ In case of spring pull and push both are possible

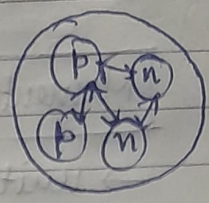
(c) Frictional Force:

→ Tangential direcⁿ of surfaces in contact.
(Shear direcⁿ)



(III) Nuclear Force:

→ Acts b/w p-p, p-n, n-n
→ Very short range force
→ Acts only if sep. is $\sim 10^{-15}$ m
$\sim$ 1 Fermi
→ It is reason for stability of nucleus



Nucleus (8×10^{-15} m)
Fermi

$\uparrow F = \frac{kQ_1Q_2}{r^2} \downarrow$

(IV) Weak Nuclear force:

$$n \longrightarrow p + e^{\ominus} + \overline{\gamma}$$

→ Neutron will decay into proton and $e^{\ominus}$ with antineutrino.
→ This disintegration is due to weak nuclear force.

 # Newton's Laws of Motion :

Newtons 1st law (Law of inertia):

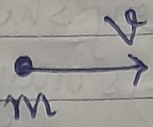
→ Inertia is property of an object due to which it has some resistance to change its motion.
→ Higher the mass, more will the body oppose to change its motion.

$$\boxed{\text{Inertia} \propto \text{mass}}$$

→ Newtons 1st law states that an object always wants to maintain its uniform motion or rest until some external force is applied to act otherwise.
→ On applying force, some objects may change their motion easily ie, they have less mass or inertia.
→ On applying force some objects may change their motion not so easily ie they have large mass or inertia.

Newtons 2nd law :

$$\boxed{\text{Momentum} = \vec{p} = m\vec{v}}$$



→ Unit = kg m/sec

→ Dimension = $[MLT^{-1}]$

→ Vector

→ Direction always along velocity.

→ Newtons 2nd law says force is requireed to change momentum and is equal to rate of change of momentum

$$\boxed{\vec{F} = \frac{d\vec{p}}{dt}} \qquad \boxed{F_{avg} = \frac{\Delta\vec{p}}{\Delta t}}$$

→ Force is vector

→ Direction always along change in momentum.

→ Unit = $kgm/s^2 = N$

→ Dimension = $[MLT^{-2}]$

Q An object of mass 2kg has velocity $\vec{v} = (2t^2\hat{i} + 5t\hat{j} + 10\hat{k})$ m/s

Find ① Force at any time t

② Force at t = 2s

③ Force at t = 5s

④ F_{Avg} in time interval t = 0s to t = 5s.

$$\vec{P} = m\vec{v} = 2 \times (2t^2\hat{i} + 5t\hat{j} + 10\hat{k})$$
$$= (4t^2\hat{i} + 10t\hat{j} + 20\hat{k}) \text{ kg m/s}$$

① $\vec{F} = \frac{d\vec{p}}{dt} = 8t\hat{i} + 10\hat{j}$ N

② $(\vec{F})_{t=2} = (16\hat{i} + 10\hat{j})$ N

③ $(\vec{F})_{t=5} = (40\hat{i} + 10\hat{j})$ N

(4) $\vec{F}_{avg} = \frac{\Delta\vec{P}}{\Delta t} = \frac{\vec{P}_5 - \vec{P}_0}{5} = \frac{(100\hat{i} + 50\hat{j} + 20\hat{k}) - (20\hat{k})}{5}$

$\vec{F}_{avg} = 20\hat{i} + 10\hat{j}$ kg m/s²

Note: $\vec{F} = \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt}$

$= m\frac{d\vec{v}}{dt} + \vec{v}\frac{dm}{dt}$ ($\frac{dm}{dt} \to 0$, m = const)

$$\boxed{\vec{F} = m\frac{d\vec{v}}{dt} = m\vec{a}}$$

$$\boxed{F_x = ma_x \;,\; F_y = ma_y \;,\; F_z = ma_z}$$

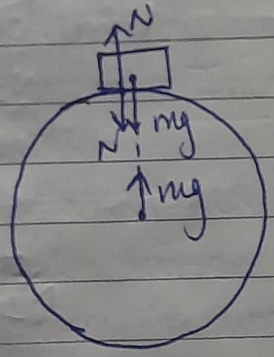
$F_H = ma_H$ $F_V = ma_V$ $F_{I.P.} = ma_{I.P.}$

Newtons 3rd law: For every action there is equal and opp. reaction

→ Action/Reaction will occur simultaneously.

→ Action/Reaction will be of same nature

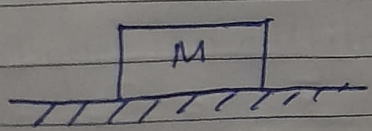
→ Action/Reaction will act on different bodies.



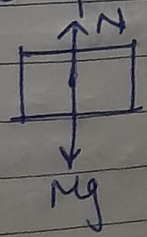
Note: N and mg are not action/reacn pairs.

Free Body diagrams (FBD): diagram which represents all forces acting on a body is known as FBD

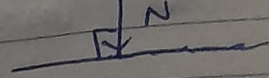
(1)



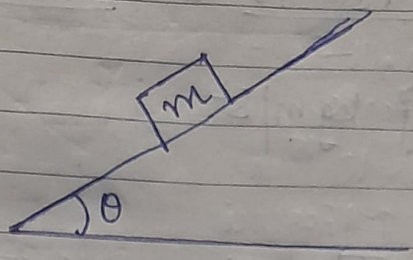
FBD of block

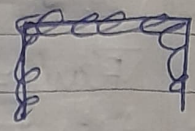


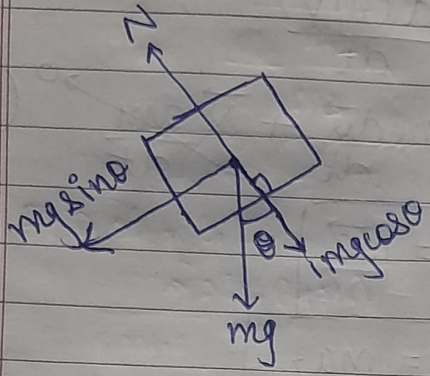
FBD of ground



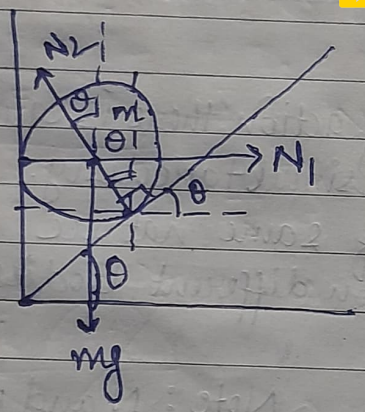
(2)







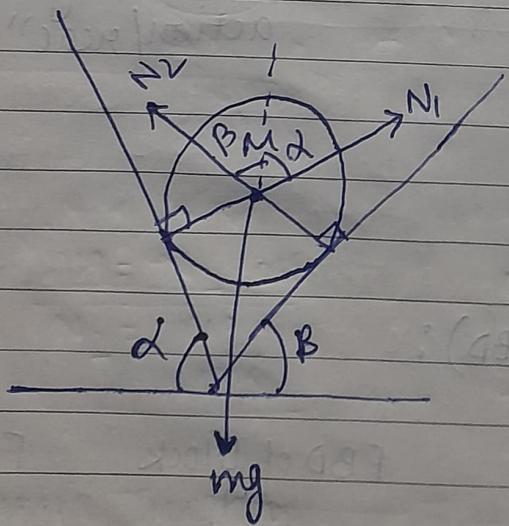
(3)

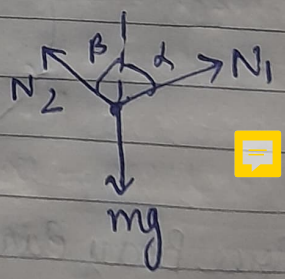


Remember:

If inclined plane is at angle θ, the N makes an angle θ with the vertical

(4)





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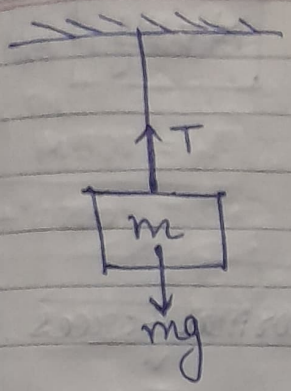
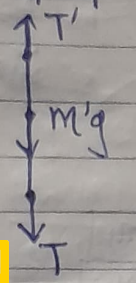
(5)

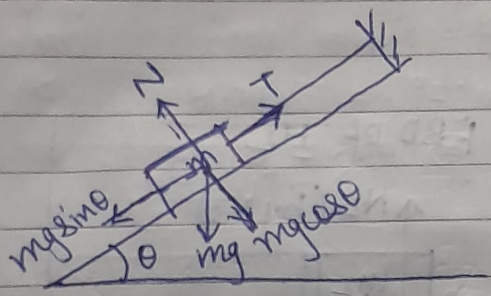


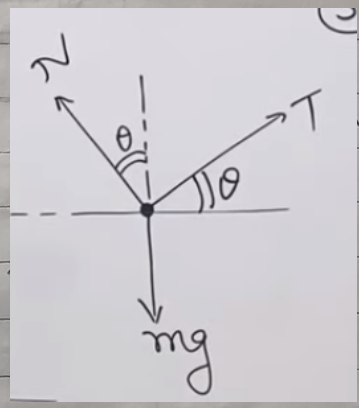
FBD of rope



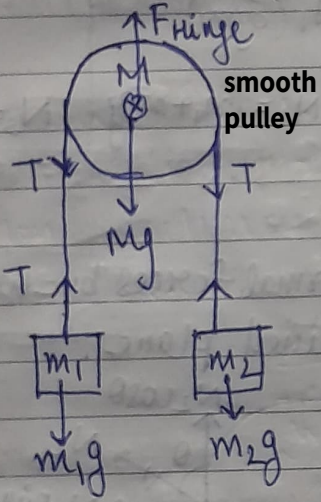
If $m' = 0$
$\Rightarrow$ massless string
$\Rightarrow$ $\boxed{T' = T}$

(6)

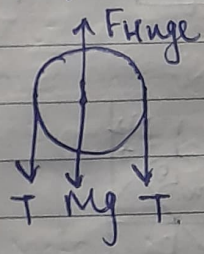




(7)



FBD of Pulley



$F_{Hinge} = Mg + 2T$

Equilibrium:

$\vec{F} = 0 \Rightarrow \vec{a} = 0 \Rightarrow \vec{v} = \text{const}$

→ Static Equilibrium $\Rightarrow \vec{v} = 0$ (Rest)

→ Dynamic Equilibrium $\Rightarrow \vec{v} = \text{const} \neq 0$ (motion)

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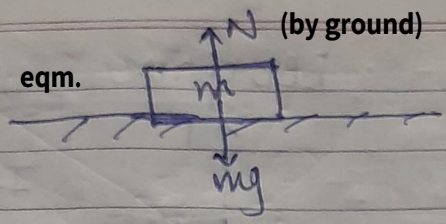
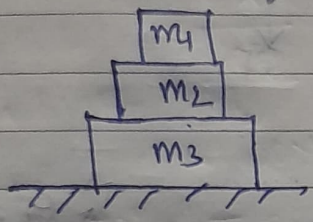


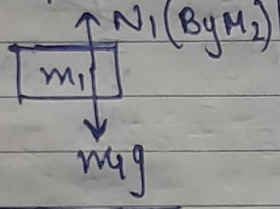
$N = mg$

Q



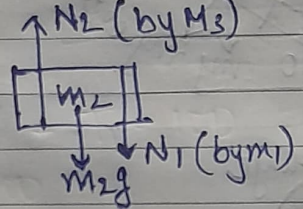
Find normal forces.

FBD of m_1



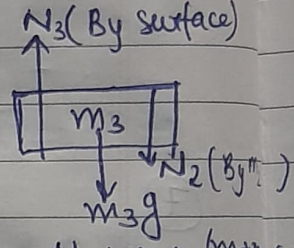
$N_1 = m_1 g$

FBD of M_2



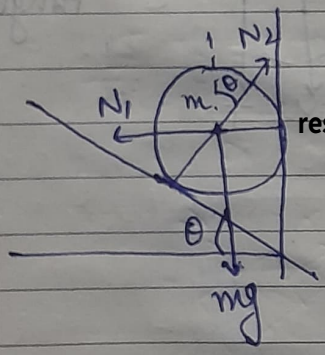
$N_2 = m_2 g + N_1 = (m_1 + m_2) g$

FBD of M_3

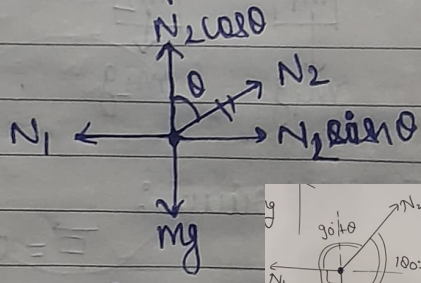


$N_3 = N_2 + m_3 g = (m_1 + m_2 + m_3) g$

Q



Find Normal forces by wall and inclined plane.

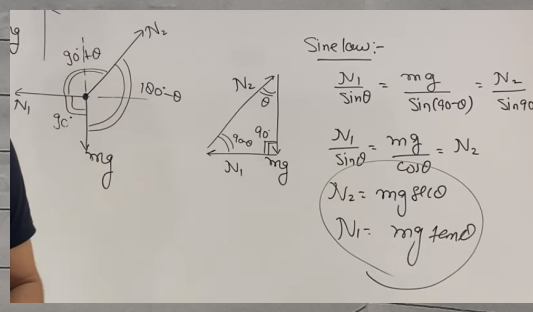


$mg = N_2 \cos\theta$

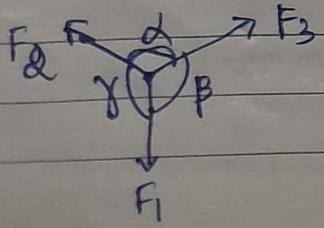
$N_2 = \dfrac{mg}{\cos\theta}$

$N_1 = N_2 \sin\theta$

$N_1 = mg \tan\theta$



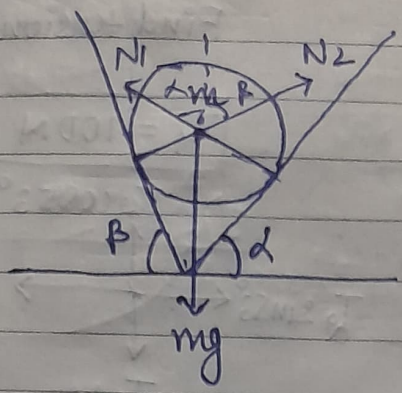
Lami's Theorem



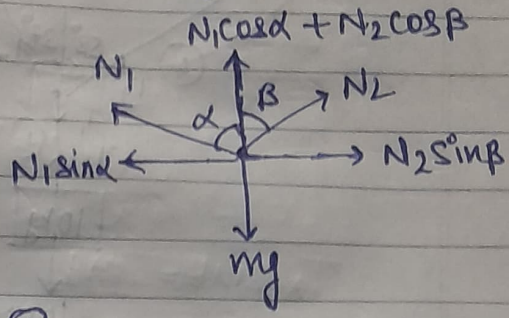
$$\Rightarrow \quad \frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$$

Note: Vector addn of F_1, F_2, F_3 must be zero.

Q



Find normal forces on the ball



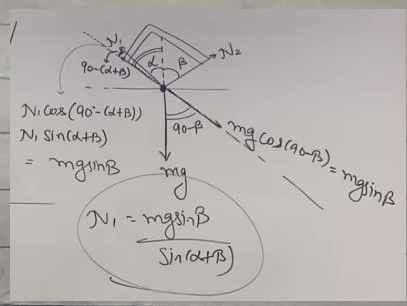
$N_1\cos\alpha + N_2\cos\beta = mg$ — (1)

$N_1\sin\alpha = N_2\sin\beta$ — (2)

$\Rightarrow N_2 = \frac{N_1\sin\alpha}{\sin\beta}$ — (3)

Put (3) in (1)

$N_1\cos\alpha + \frac{N_1\sin\alpha}{\sin\beta}\cos\beta = mg$



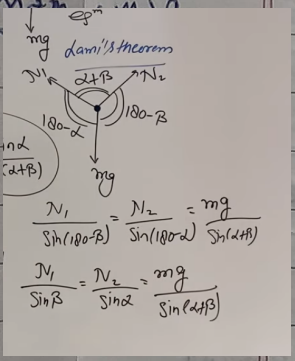
$= N_1\cos\alpha + N_1\sin\alpha\cot\beta = mg \Rightarrow N_1\left(\frac{\cos\alpha\sin\beta + \sin\alpha\cos\beta}{\sin\beta}\right) = mg$

~~$N_1(\cos\alpha + \sin\alpha$~~

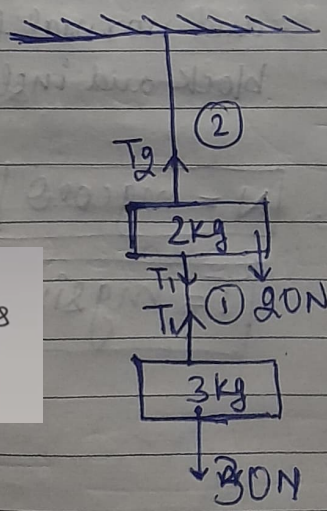
$= N_1\left(\frac{\sin(\alpha+\beta)}{\sin\beta}\right) = mg$

$= \boxed{N_1 = \frac{mg\sin\beta}{\sin(\alpha+\beta)}}$

$\boxed{N_2 = \frac{mg\sin\alpha}{\sin(\alpha+\beta)}}$



Q



Find tension in ① and ②
(Strigs are ideal/massless)

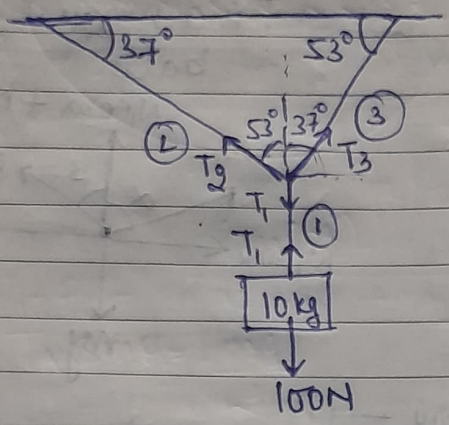
$\boxed{T_1 = 30N}$

$T_2 = 20 + 30$

$\boxed{T_2 = 50N}$

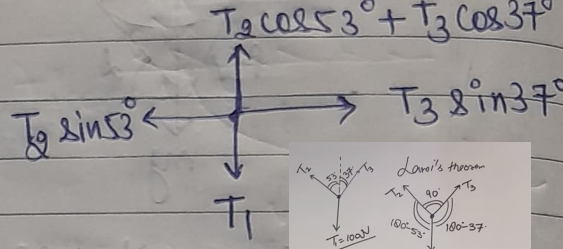
ideal strings
massless strings
light strings.

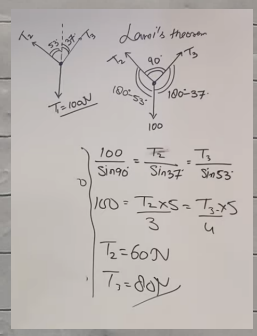
Q



Find tension in all string

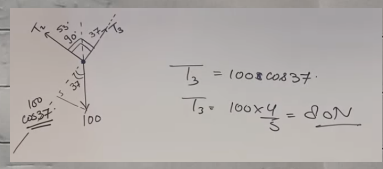
$T_1 = 100N$





$T_2 \sin 53° = T_3 \sin 37°$

$= \frac{4T_2}{5} = \frac{3}{5} T_3 \Rightarrow \boxed{T_2 = \frac{3}{4} T_3}$



$T_2 \cos 53° + T_3 \cos 37° = T_1 = 100$

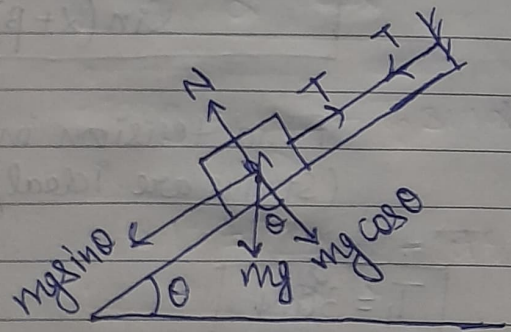
$= \frac{3}{5} T_2 + \frac{4}{5} T_3 = 100$

$= \frac{3}{5} \times \frac{3}{4} T_3 + \frac{4}{5} T_3 = 100 \Rightarrow \frac{9}{20} T_3 + \frac{4}{5} T_3 = 100$

$= \frac{5}{4} T_3 = 100 \Rightarrow \boxed{T_3 = 80N}$

$\boxed{T_2 = 60N}$

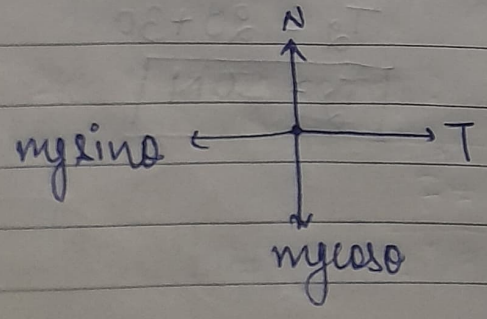
Q

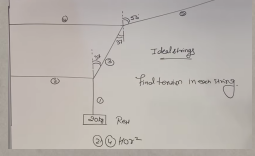


Find tension in string and normal force b/w block and inclined plane

$\boxed{N = mg\cos\theta}$

$\boxed{T = mg\sin\theta}$





Page No.
Date:

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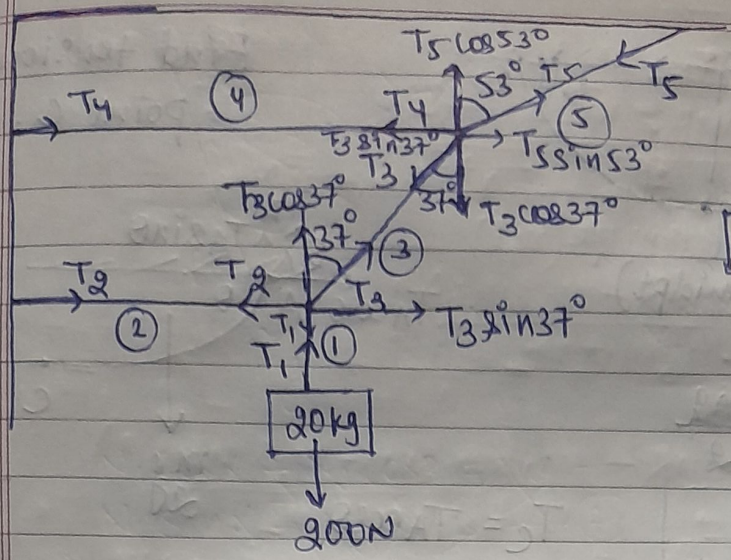


Find tension in each string.

$T_1 = 200N$

$T_3\cos37° = 200$

$T_3 = \frac{\cancel{200}^{50} \times 5}{\cancel{4}}$

$T_3 = 250N$

$T_2 = T_3\sin37°$

$T_2 = 150N$

$T_5\cos53° = T_3\cos37°$

$T_5 = 200 \times \frac{5}{5}$

~~$T_5 = 200N$~~

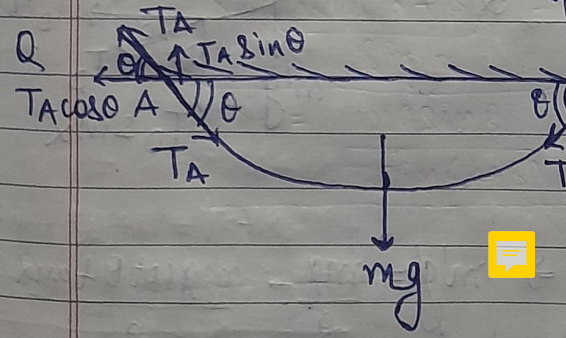
$T_5 = \frac{1000}{3} N$

$T_4 = T_5\sin53° - T_3\sin37°$

$T_4 = \frac{4}{5} \times \frac{\overset{200}{\cancel{1000}}}{3} - 150$

$T_4 = \frac{350}{3} N$

Q



It is a uniform rope of mass m
Find tension in rope at A & B

$T_A\cos\theta = T_B\cos\theta$

$T_A = T_B$

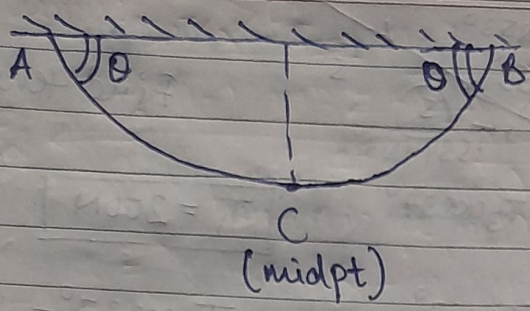
$T_A\sin\theta + T_B\sin\theta = mg$

$2T_A\sin\theta = mg$

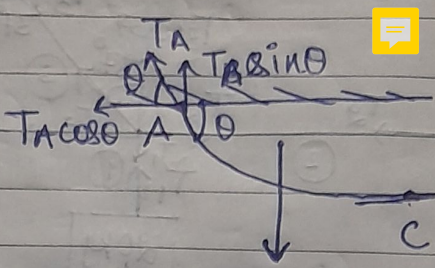
$T_A = \frac{mg}{2\sin\theta} = T_B$

Page No.
Date:

Q



Find tension at point C.



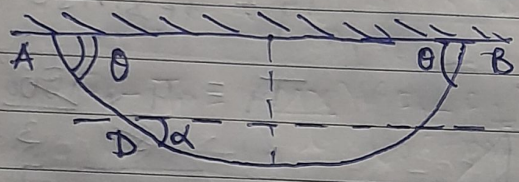
$T_A \sin\theta = \frac{mg}{2}$

$$\boxed{T_A = \frac{mg}{2\sin\theta}}$$

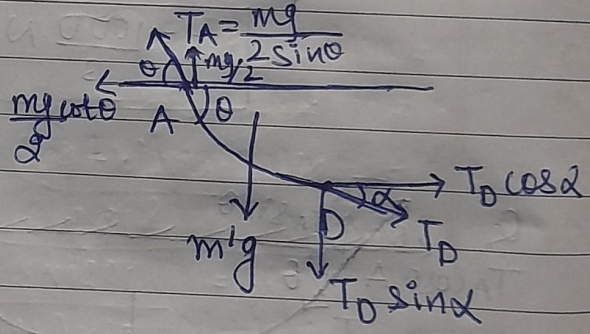
$T_C = T_A \cos\theta$

$$\boxed{T_C = \frac{mg\cot\theta}{2}}$$

Q



Find tension at pt. D



$T_D \cos\alpha = \frac{mg\cot\theta}{2}$

$$\boxed{T_D = \frac{mg\cot\theta}{2\cos\alpha}}$$

$m'g = \frac{mg}{2} - T_D \sin\alpha \Rightarrow m'g = \frac{mg}{2} - \frac{mg\cot\theta\tan\alpha}{2}$

$$\boxed{m' = \frac{m}{2}(1 - \cot\theta\tan\alpha)}$$

Accelerating bodies:

$$\boxed{\vec{F} = m\vec{a}}$$

$f_x = m a_x$

$f_H = m a_H$

$f_{I.P} = m a_{I.P}$

Page No.
Date:

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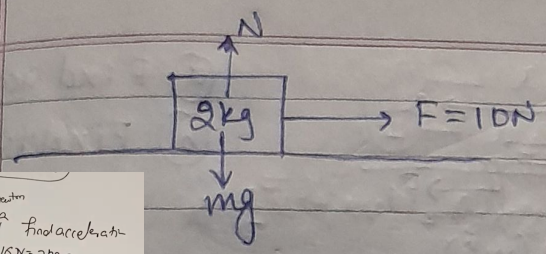
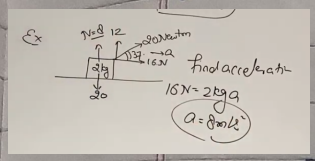


Find acc^n

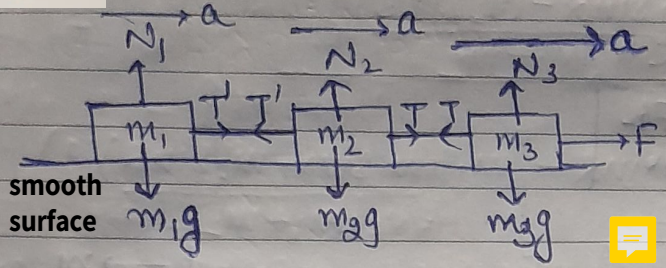
$$a = \frac{F}{m} = \frac{10}{2} = 5\,m/s^2$$



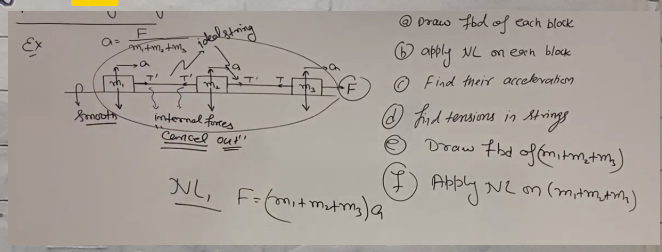
Q

ideal strings

smooth surface



(a) Draw FBD of each block
(b) Apply NL on each block
(c) Find acc^n of blocks
(d) Find tension in strings



$$F - T = m_3 a \quad -(1)$$
$$T - T' = m_2 a \quad -(2)$$
$$+\ T' = m_1 a \quad -(3)$$

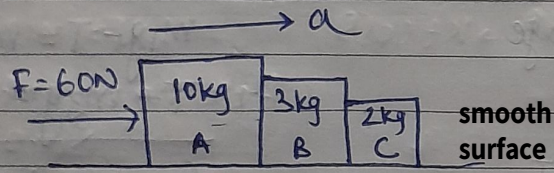
$$F = (m_1 + m_2 + m_3)a$$

$$\boxed{a = \frac{F}{m_1 + m_2 + m_3}}$$

$$\boxed{T' = \frac{m_1 F}{m_1 + m_2 + m_3}}$$

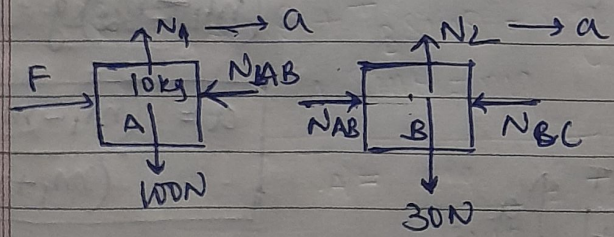
$$\boxed{T = \frac{(m_2 + m_1) F}{(m_1 + m_2 + m_3)}}$$

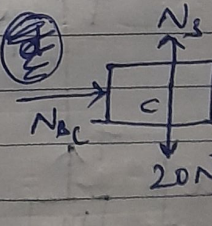
Q

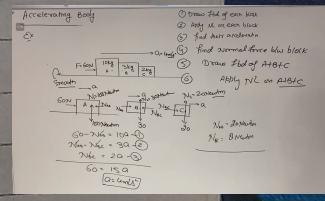


(a) Draw FBD of each block
(b) Find their acc^n
(c) Find Normal reac^n b/w blocks

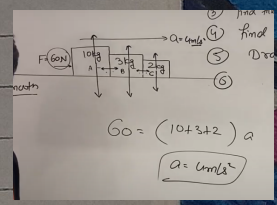
(a)







(b) $$a = \frac{F}{m_A + m_B + m_C} = \frac{60}{15} = 4\,m/s^2$$



(c) $60 - N_{AB} = 10 \times 4$

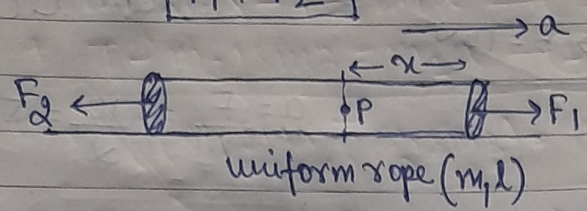
$$\boxed{N_{AB} = 20N}$$

$N_{AB} - N_{BC} = 3 \times 4$

$$\boxed{N_{BC} = 8N}$$

Q Given: $F_1 > F_2$

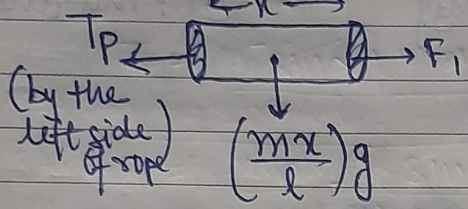
(a) Find acc^n of rope
(b) Find tension at pt P.



(a) $F_1 - F_2 = ma$

$$a = \frac{F_1 - F_2}{m}$$

(b)



$l \longrightarrow m$

$1 \longrightarrow \frac{m}{l}$

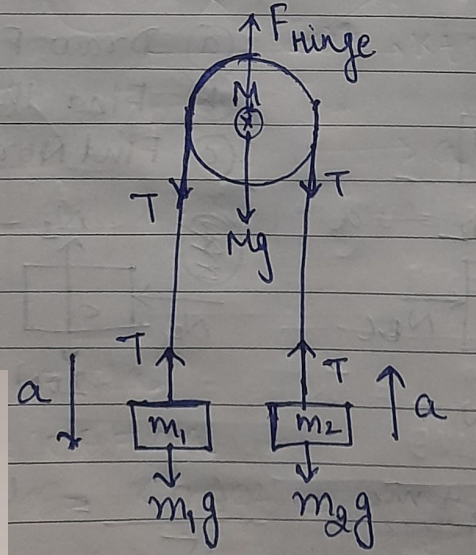
$x \longrightarrow \frac{mx}{l}$

$$F_1 - T_P = \left(\frac{mx}{l}\right)\left(\frac{F_1 - F_2}{m}\right)$$

$$T_P = F_1 - (F_1 - F_2)\left(\frac{x}{l}\right)$$

Atwood Machine:

$m_1 > m_2$

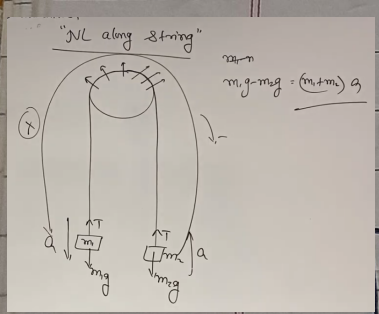


$$m_1 g - T = m_1 a \quad (1)$$
$$+\ T - m_2 g = m_2 a \quad (2)$$
$$(m_1 + m_2) g = (m_1 + m_2) a$$

$$a = \frac{(m_1 - m_2) g}{(m_1 + m_2)}$$

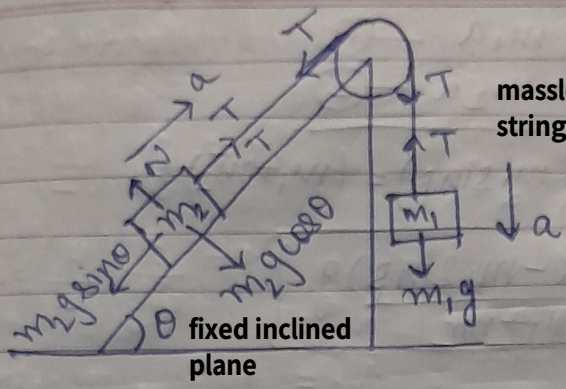
$$T = \frac{2 m_1 m_2 g}{m_1 + m_2}$$

$$F_{Hinge} = Mg + 2T = Mg + \frac{4 m_1 m_2 g}{m_1 + m_2}$$



$m_1 > m_2$

Q



Find acc^n of blocks and tension in strings.

$m_1 g - T = m_1 a$ — ①

$+ \quad T - m_2 g \sin\theta = m_2 a$ — ②

$g(m_1 - m_2 \sin\theta) = (m_1 + m_2) a$

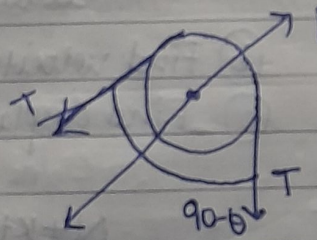
$$a = \frac{(m_1 - m_2 \sin\theta)\, g}{m_1 + m_2}$$

$$T = m_1 g - \left\{ \frac{(m_1 - m_2 \sin\theta) g}{m_1 + m_2} \right\}$$

$$T = \frac{(m_1^2 + m_1 m_2 - m_1^2 + m_1 m_2 \sin\theta) g}{m_1 + m_2}$$

$$T = \frac{m_1 m_2 g (1 + \sin\theta)}{m_1 + m_2}$$

Also Find F_{Hinge} on pulley

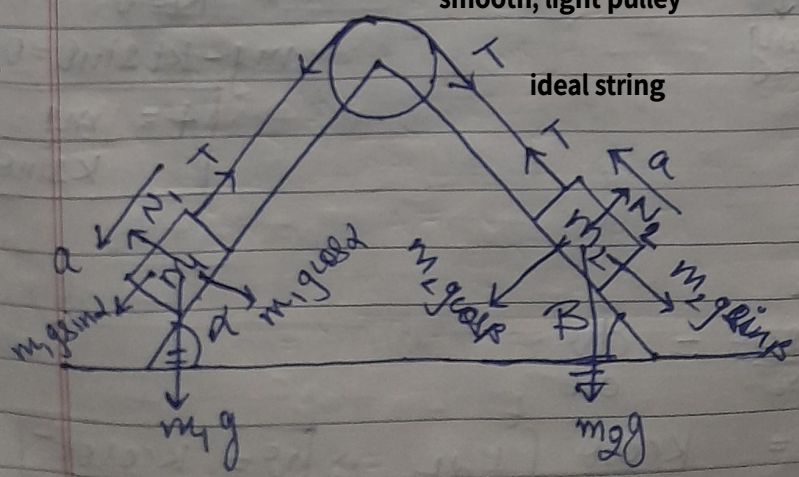


$$\sqrt{T^2 + T^2 + 2T^2 \cos(90 - \theta)}$$

$$= \sqrt{2T^2 (1 + \sin\theta)}$$

$$\therefore F_{Hinge} = T\sqrt{2}\sqrt{(1 + \sin\theta)}$$

Q



Find acc^n of each block and tension in string.

$$m_1 g\sin\alpha - T = m_1 a \quad \text{—(1)}$$
$$+\ T - m_2 g\sin\beta = m_2 a \quad \text{—(2)}$$
$$g(m_1\sin\alpha - m_2\sin\beta) = (m_1+m_2)a$$

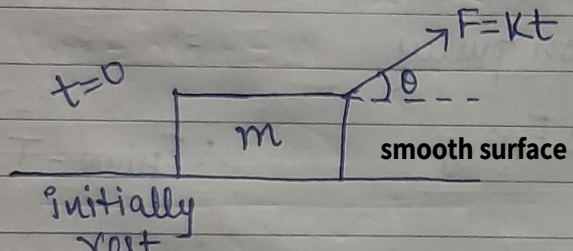
$$\boxed{a = \frac{(m_1\sin\alpha - m_2\sin\beta)g}{m_1+m_2}}$$

$$T = m_1 g\sin\alpha - m_1\left(\frac{(m_1\sin\alpha - m_2\sin\beta)g}{m_1+m_2}\right)$$

$$T = \left(\frac{m_1^2\sin\alpha + m_1 m_2\sin\alpha - m_1^2\sin\alpha + m_1 m_2\sin\beta}{m_1+m_2}\right)g$$

$$\boxed{T = \frac{m_1 m_2 g(\sin\alpha+\sin\beta)}{m_1+m_2}}$$

Q

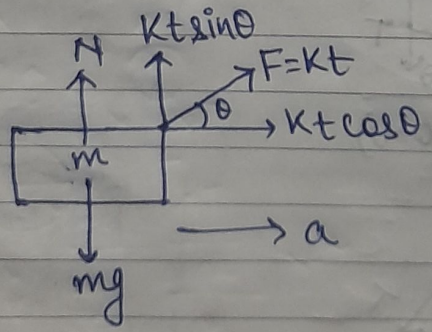


(a) Find the time when block leave off the ground

(b) Find velocity of block at this time

(c) Find displacement of block at this time

(a)



$$N + Kt\sin\theta = mg$$
$$N = mg - Kt\sin\theta$$

For no contact from ground

$$N = 0$$
$$mg - Kt\sin\theta = 0$$
$$\boxed{t = \frac{mg}{K\sin\theta}}$$

(b)

$$F_x = ma$$
$$Kt\cos\theta = ma$$
$$a = \frac{K\cos\theta\, t}{m} = \frac{dv}{dt}$$
$$\int_0^v dv = \frac{K\cos\theta}{m}\int_0^t t\,dt \Rightarrow \boxed{v = \frac{K\cos\theta\, t^2}{2m}}$$

Velocity at ~~time at~~ which it leaves the ground = $\frac{K\cos\theta}{2m}\left(\frac{m^2g^2}{K^2\sin^2\theta}\right)$

$$\boxed{v' = \frac{mg^2\cos\theta}{2K\sin^2\theta}}$$

(c) $v = \frac{dx}{dt}$

$$\int_0^x dx = \frac{K\cos\theta}{2m}\int_0^t t^2\,dt \Rightarrow \boxed{x = \frac{K\cos\theta\, t^3}{6m}}$$

Displacement at time at which it leaves ground =

$$\boxed{x' = \frac{K\cos\theta}{6m}\left(\frac{m^3g^3}{K^3\sin^3\theta}\right) = \frac{m^2g^3\cos\theta}{6K^2\sin^3\theta}}$$

Constraint Motion:

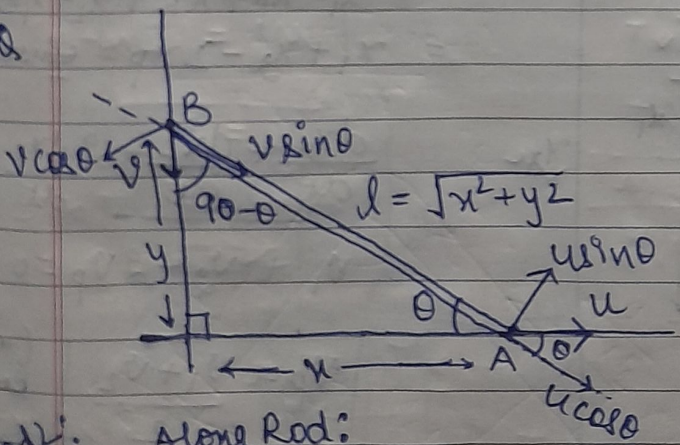
→ Motion of two or more bodies are interdependent

mathematical relation exists b/w velocity, acceleration etc. of different bodies

Rod Constraint:

→ Length of Rod is always fixed.

Q



Find velocity of end B.

Method 1: $x^2+y^2 = l^2 = \text{const}$

diff wrt t

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

$$xu + y(-v) = 0$$

$$\boxed{v = \frac{x}{y}u = u\cot\theta}$$

Method 2: Along Rod:

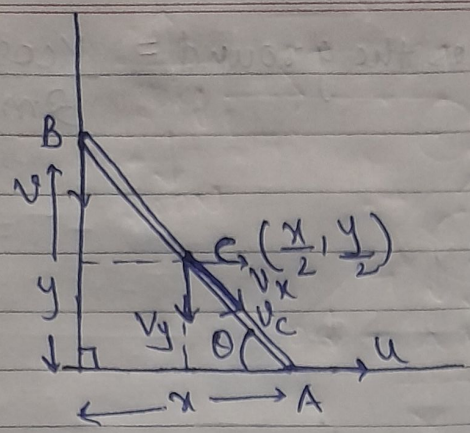
$v_{app} = v\sin\theta - u\cos\theta$

$0 = v\sin\theta - u\cos\theta$

$v\sin\theta = u\cos\theta$

$$\boxed{v = u\cot\theta}$$

Q



Find velocity of midpt C

$$v_x = \frac{d}{dt}\left(\frac{x}{2}\right) = \frac{1}{2}\frac{dx}{dt} = \frac{u}{2}$$

$$v_y = \frac{d}{dt}\left(\frac{y}{2}\right) = \frac{1}{2}\frac{dy}{dt} = \frac{-v}{2} = \frac{-u}{2}\cot\theta$$

$$\vec{v_c} = \frac{u}{2}\hat{i} - \frac{u}{2}\cot\theta\,\hat{j}$$

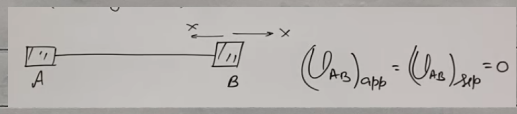
$$|\vec{v_c}| = \frac{u}{2}\sqrt{1+\cot^2\theta} = \frac{u}{2}\operatorname{cosec}\theta$$

String Constraint:

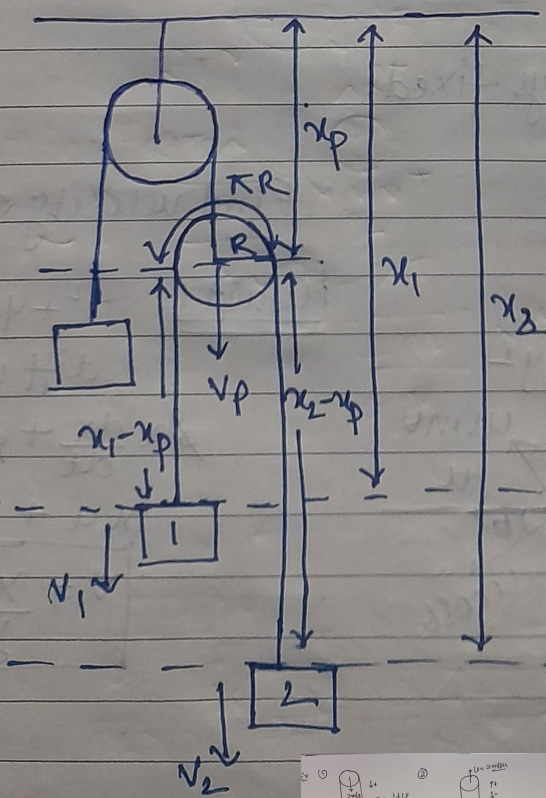
→ String doesn't break
→ String is inextinsible
→ String will not stack (always taut)

} Length of string is fixed

→ $v_{app} = v_{sep} = 0$



Ex



Length of string

$= x_1 - x_p + \pi R + x_2 - x_1 = l = \text{const}$

$= x_1 + x_2 - 2x_p = \text{const}$

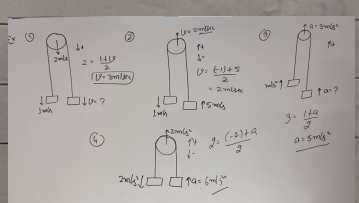
$$\frac{dx_1}{dt} + \frac{dx_2}{dt} - 2\frac{dx_p}{dt} = 0$$

$$v_1 + v_2 - 2v_p = 0$$

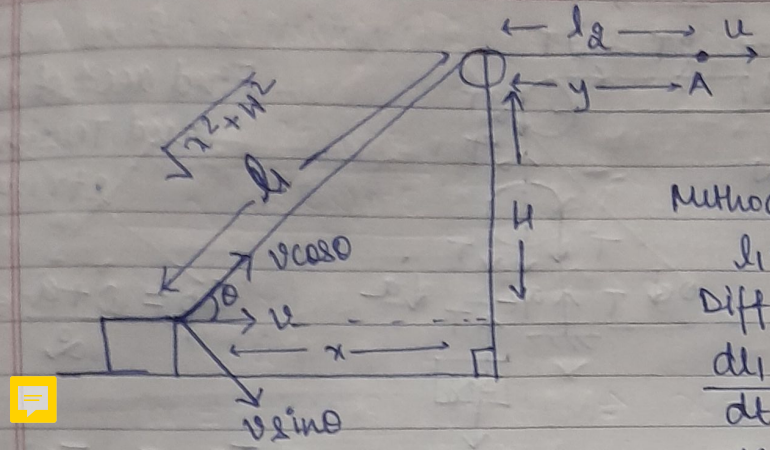
$$\boxed{v_p = \frac{v_1 + v_2}{2}}$$

(with Ro direc

$$\boxed{a_p = \frac{a_1 + a_2}{2}}$$



Q



Find velocity of block.

Method 1:

$l_1 + l_2 = \text{const} = l$

Diff wrt t

$$\frac{dl_1}{dt} + \frac{dl_2}{dt} = 0$$

$$= -v\cos\theta + u = 0$$

$$\boxed{v = u\sec\theta}$$

Method 2:

$\sqrt{x^2+h^2} + y = \text{const}$ (length of string)

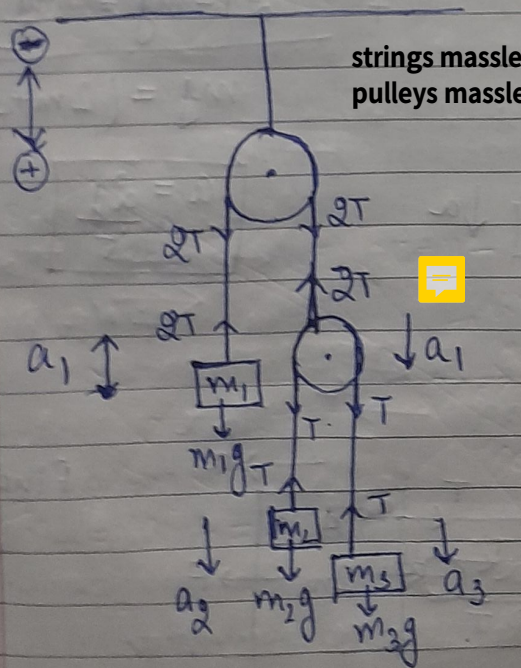
Diff wrt t

$$\frac{1}{2}\frac{2x}{\sqrt{x^2+h^2}}\left(\frac{dx}{dt}\right) + \frac{dy}{dt} = 0$$

$$= \frac{x}{\sqrt{x^2+h^2}}(-v) + u = 0$$

$$= v = \frac{\sqrt{x^2+h^2}}{x} \times u = u\sec\theta$$

Q



strings massless
pulleys massless, smooth

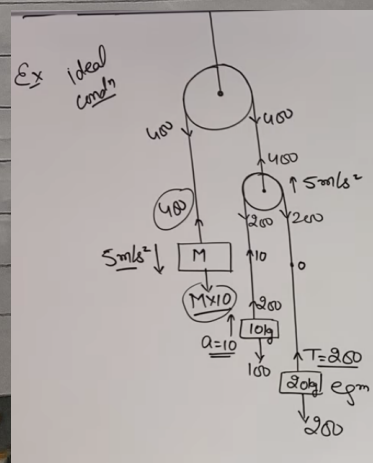
$2T - m_1g = m_1a_1$ — (1)

$m_2g - T = m_2a_2$ — (2)

$m_3g - T = m_3a_3$ — (3)

$$a_1 = \frac{a_2 + a_3}{2}$$ — (4)

Ex ideal condn



Find M

Such that 20kg block is eq at eqm

$200 - 100 = 100 = 10a$

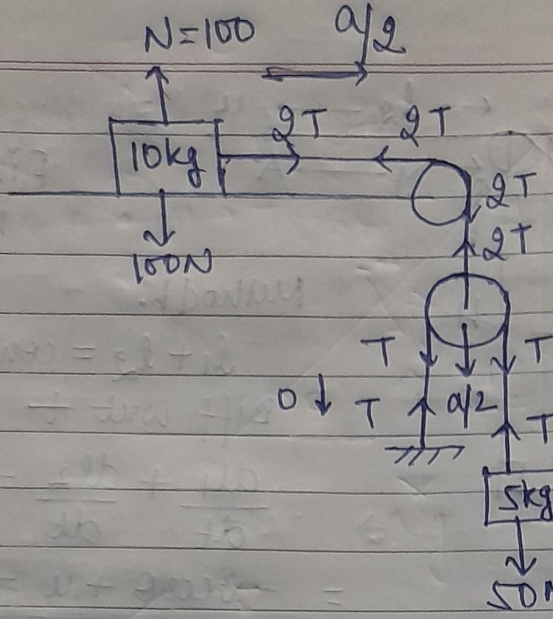
$a = 10\text{m/s}^2$

$10M - 400 = M \times 5$

$5M = 400$

$M = 80\text{kg}$

Q



ideal pulley

Find acc^n of blocks and Tension in strings.

$a_p = \frac{0+a}{2} = \frac{a}{2}$

$2(50 - T = 5a)$ — ①

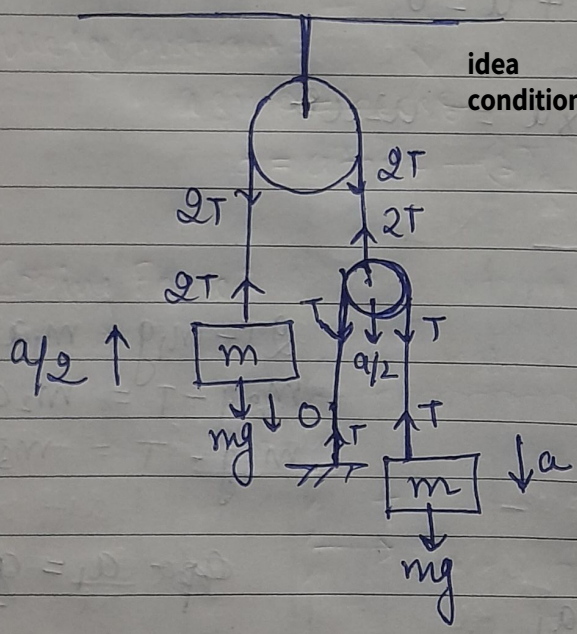
$2T = 10\frac{a}{2}$ — ②

$100 = 15a$

$a = \frac{100}{15} = \frac{20}{3}\ m/s^2$

$T = \frac{5a}{2} = \frac{5}{2} \times \frac{20}{3} = \frac{50}{3} N$

Q



idea conditions

Find acc^n of the blocks

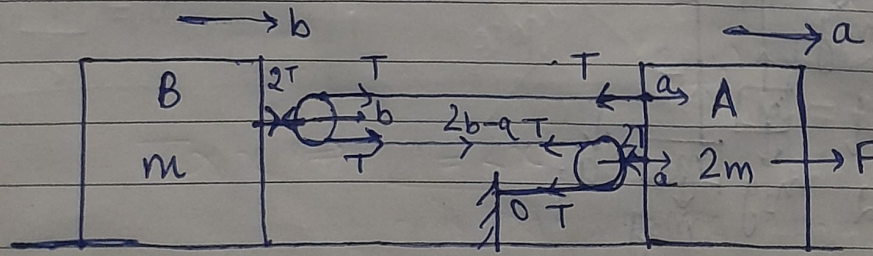
$2(mg - T = ma)$ — ①

$2T - mg = \frac{ma}{2}$ — ②

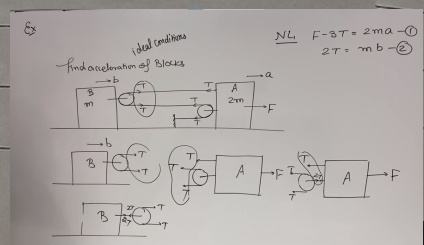
$mg = (2m + \frac{m}{2})a$

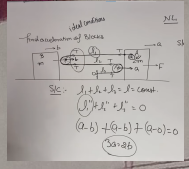
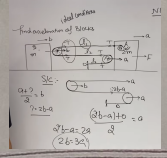
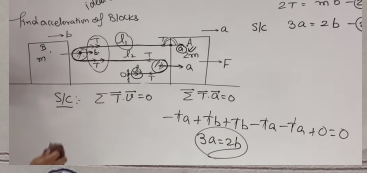
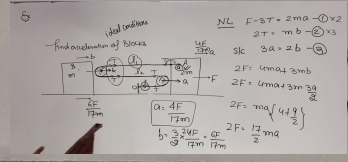
$a = \frac{2g}{5}$

Q



Find acc^n of blocks



$F - 3T = 2ma$ — (1)

$2T = mb$ — (2)

String const: $2b - a = 2a$

$2b = 3a$ — (3)

$b = \frac{6F}{17m}$ $\quad a = \frac{4F}{17m}$

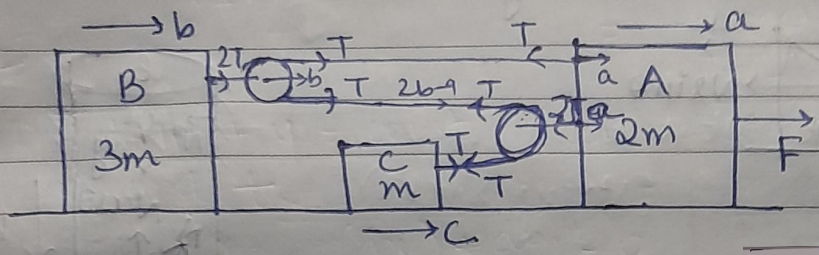
Note:

(1) $\sum \vec{T} \cdot \vec{v} = 0$

(2) $\sum \vec{T} \cdot \vec{a} = 0$

These can also be used to find relations

Q



Find acc^n of blocks

$F - 3T = 2ma$ — (1)

$2T = 3mb$ — (2)

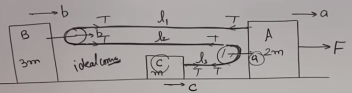
$T = mc$ — (3)

String Constraint:

~~$2b + 3a = 5a$~~ ~~(4)~~

$2b + c = 3a$ — (4)

Ex



Find accelerations of blocks.

N.L. $F - 3T = (2m)a$ — (1)

$2T = (3m)b$ — (2)

$T = mc$ — (3)

$3a = 2b + c$ — (4)

S/C $l_1 + l_2 + l_3 = C$

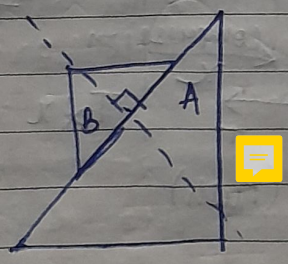
$l_1'' + l_2'' + l_3'' = 0$

$(a-b) + (a-b) + (a-c) = 0$

Wedge Constraint: wedge = blocks

→ Wedge are rigid ie shape, size, geometry are unchanged

→ Wedges will remain in contact



No relative motion is allowed along line ⊥ to common surface

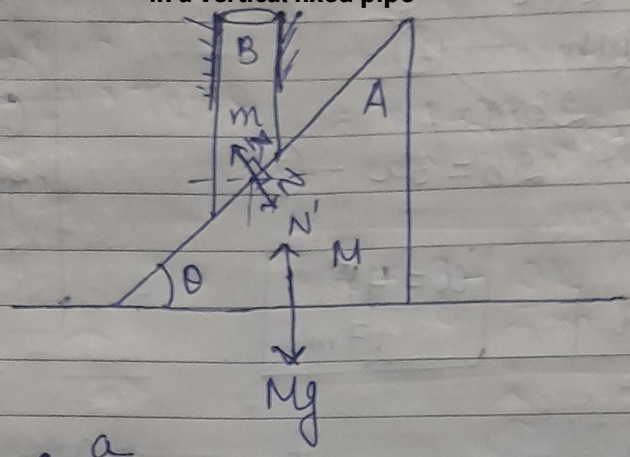
$\Rightarrow x_{rel} = 0,\ v_{rel} = 0,\ a_{rel} = 0$

Relative motion is allowed only along common surface

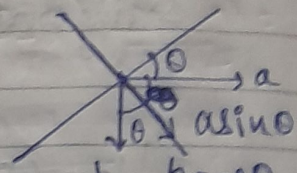
∴ Along normal to common surface both wedges will have same velocity or acc^n.

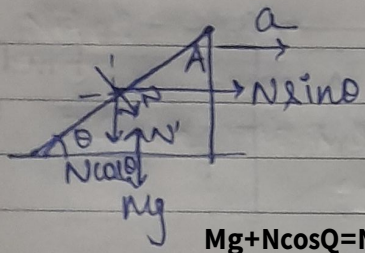
Along normal to common surface.
- their will be no Relative motion
- Both blocks will have same velocity
- Both blocks will have same acceleration

block B is fixed
in a vertical fixed pipe



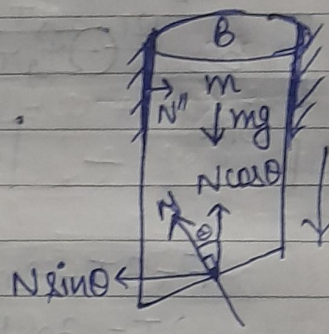
Find acc^n of blocks.





Mg+NcosQ=N'

$N\sin\theta = Ma$ — (1)



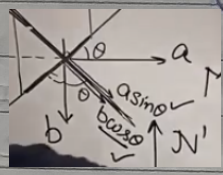
$N'' = N\sin\theta$

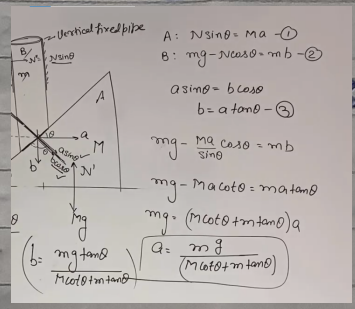
$mg - N\cos\theta = mb$ — (2)

Wedge Constraint:

$a\sin\theta = b\cos\theta$

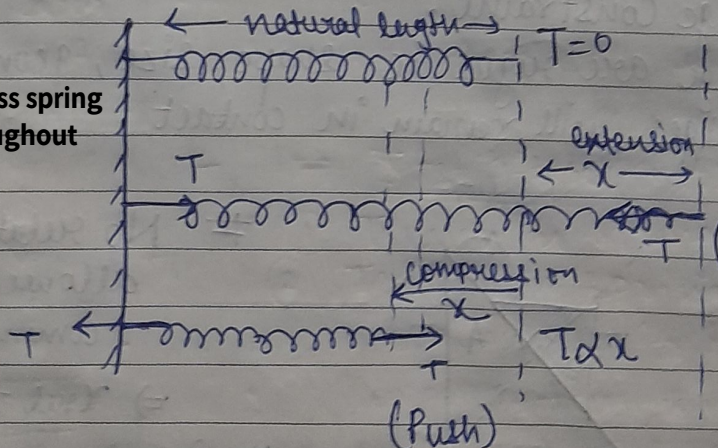
$b = a\tan\theta$ — (3)





Spring Force:

ideal spring : massless spring
hence, tension throughout the spring same



$T \propto x \Rightarrow \boxed{T = kx}$

x is extension or compression :

or stiffness constant

K = spring constant (scalar)

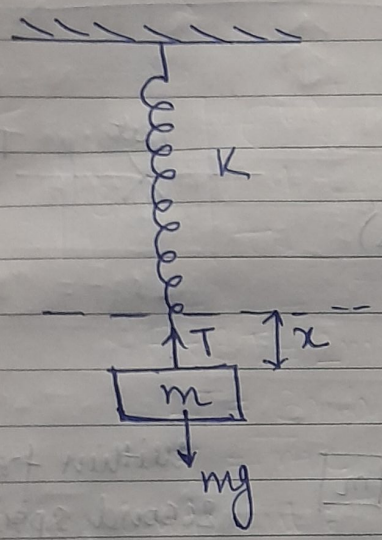
→ It is a property of spring

→ $K \propto \frac{1}{\text{natural length}}$

→ Unit : N/m

→ Dimension : $[MT^{-2}]$

Q

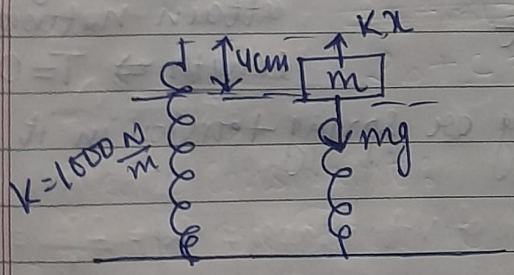


Find extension in spring.

$Kx = mg$

$$x = \frac{mg}{K}$$

Q

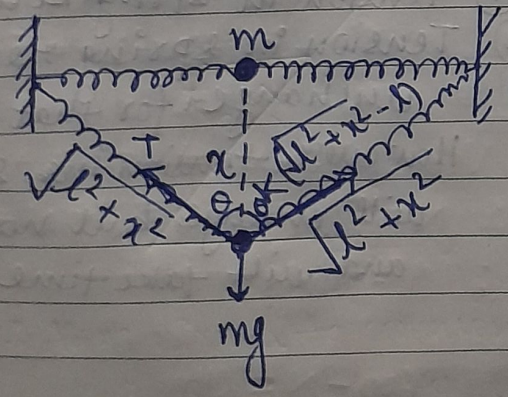


Find m such that it is in equilibrium.

$Kx = mg$

$$m = \frac{Kx}{g} = \frac{1000 \times \frac{4}{100}}{10} = 4\,kg$$

Q

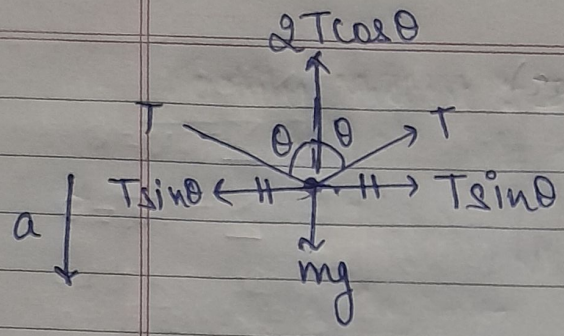


Release

Given: Spring constant = k
natural length = l

Find acceleration of m when it has descended by distance x.

Page No.
Date:



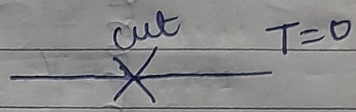
$$mg - 2T\cos\theta = ma$$

$$= mg - 2K\left(\sqrt{l^2+x^2} - l\right)\left(\frac{x}{\sqrt{l^2+x^2}}\right) = ma$$

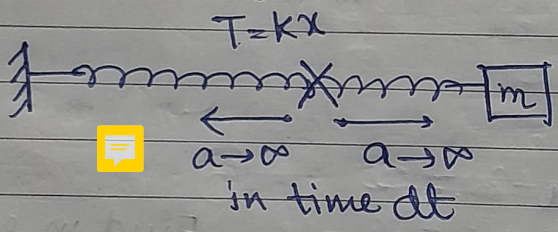
$$\boxed{a = g - \frac{2K}{m}\left(\sqrt{l^2+x^2} - l\right)\left(\frac{x}{\sqrt{l^2+x^2}}\right)}$$

Note:

(1) If we cut a string $T=0$ (string slacks)

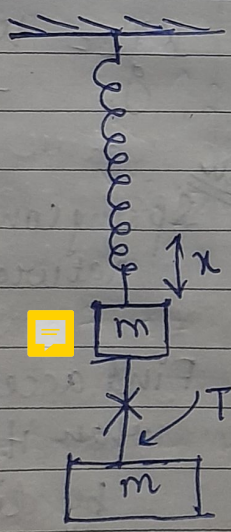


(2) In case of spring:



within fraction of second spring will attain natural length ⇒ $T=0$

∴ Just after cutting a string or spring tension in it will become zero



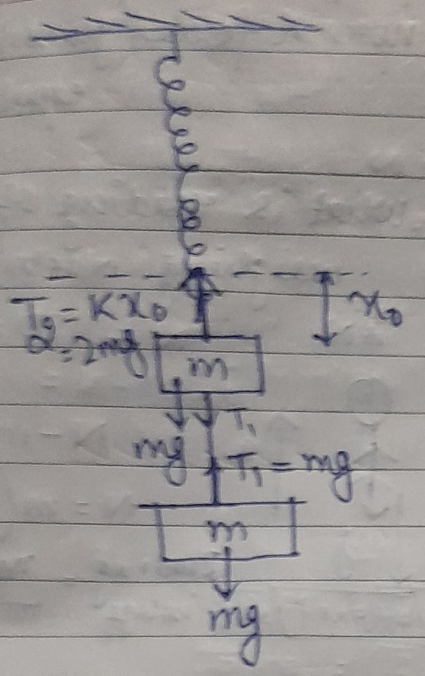
⇒ Just after cutting string:

(1) Tension in string becomes zero

(2) Tension in spring remains unchanged for that instant. However, tension in spring will change but not instantly and will take time.

Page No.
Date:

8



System is in rest.

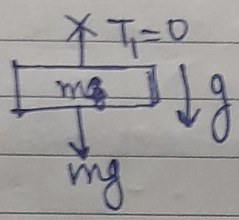
① Find tension in spring and string.

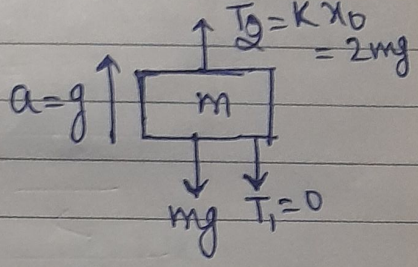
② Find accelerations of blocks just after cutting the string.

① $T_1 = mg$

② $T_2 = Kx_0 = 2mg$

② Just after cutting string:

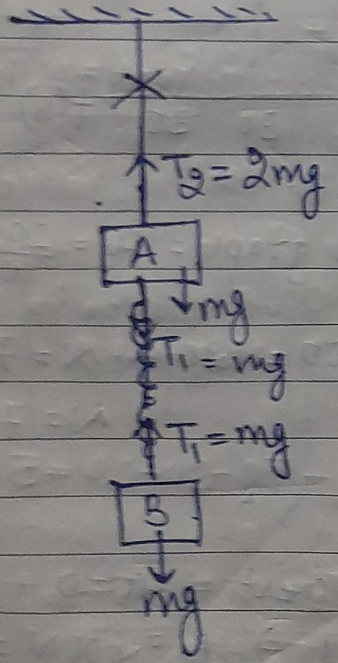




$2mg - mg = ma$

$\boxed{a = g}$

9



System is in rest.

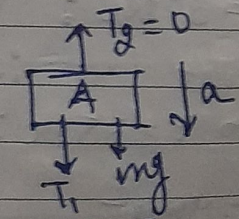
① Find tension in spring and string.

② Find acceleration of blocks just after cutting the string.

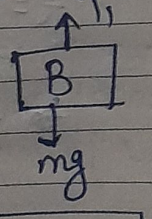
① $T_1 = mg$

$T_2 = T_1 + mg = 2mg$

② Just after cutting string:



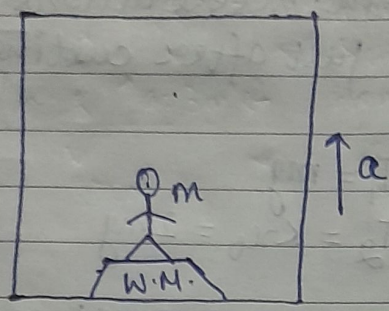
$\boxed{a = 2g}$



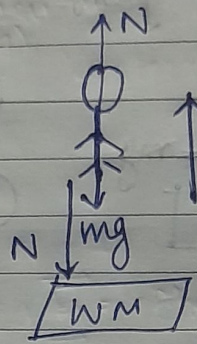
$\boxed{a = 0}$

Weighing Machine : It measures normal force (generally N is equal to wt.)

Q



What is reading of weighing machine?

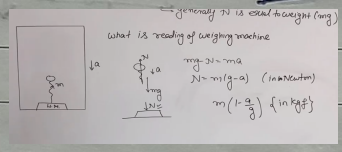


$N - mg = ma$

$N = m(g+a)$ Newton

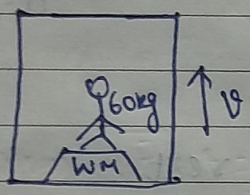
$\text{Reading in kgf} = \frac{m(g+a)}{g}$ kgf

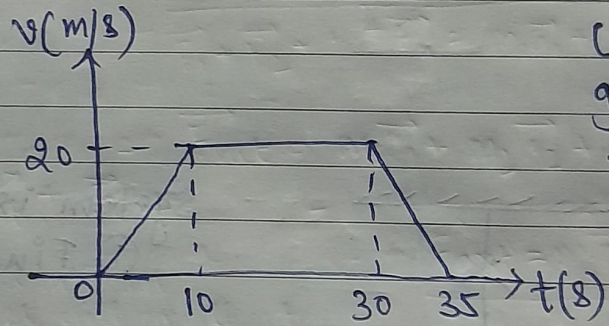
kgf = kilogram force



1 kgf = 9.8 Newton

Q





Given v v/s t graph, draw reading of W.M v/s t graph.

Q

$t = 0$ sec to $t = 10$ sec : Slope of v-t graph = $a = \frac{20}{10} = 2 m/s^2$

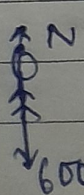
$\Rightarrow N - 600 = 120 \Rightarrow N = 720N$

$N = 72$ kgf

$t = 10$ sec to $t = 30$ sec : Velocity const $\Rightarrow a = 0$

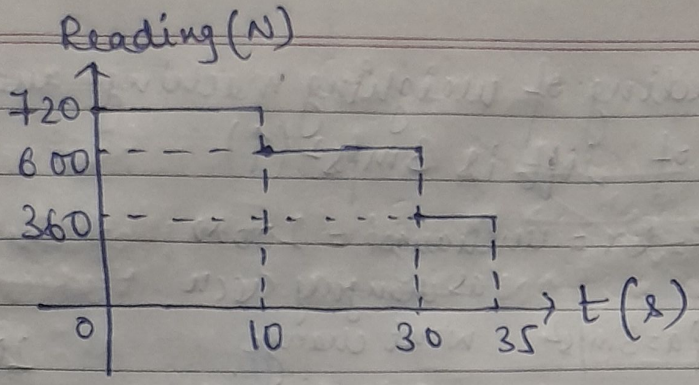
$\therefore N = mg \Rightarrow N = 600N = 60$ kgf

$t = 30$ sec to $t = 35$ sec : $a = -4 m/s^2$

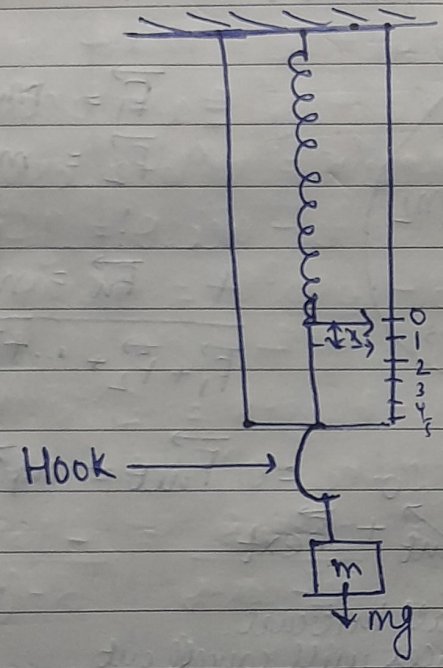


$600 - N = 240$

$N = 360N = 36$ kgf



Spring Balance: It is a spring based system which measures tension in it.



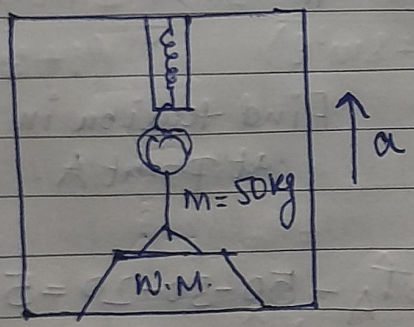
$T = Kx$

$\Rightarrow mg = T$

$mg = Kx$

$m = \frac{Kx}{g}$

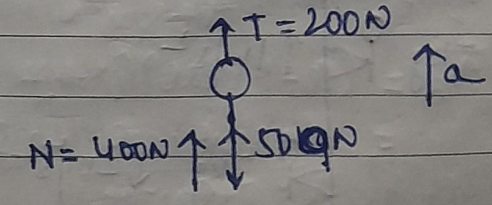
Q



Reading of W.M = 40kg

Reading of S.B. = 200kg

① Find acceleration of lift.



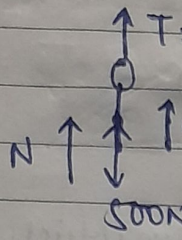
$\Rightarrow 200 + 400 - 500 = 50a$

$100 = 50a$

$\boxed{a = 2 m/s^2}$

Neelgagan
Page No.
Date:

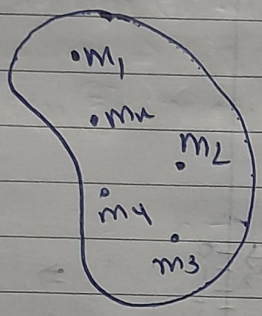
② Find reading of weighing machine and spring balance if accn of lift is $3 m/s^2 (\uparrow)$



$200 + N - 500 = 150$

$N = 450 = 45 kgf$

Newtons Law on system :



$\vec{F_1} = m_1 \vec{a_1}$

$\vec{F_2} = m_2 \vec{a_2}$

$\vdots$

$+ \quad \vec{F_n} = m_n \vec{a_n}$

$\vec{F_1} + \vec{F_2} + \dots + \vec{F_n} = m_1 \vec{a_1} + m_2 \vec{a_2} + \dots + m_n \vec{a_n}$

$= \vec{F_{net}} = m_1 \vec{a_1} + m_2 \vec{a_2} + \dots + m_n \vec{a_n}$

Now, $\vec{F_{net}} = \vec{F_{int}}^{\,0} + \vec{F_{ext}}$

(As all internal forces will cancell out as same magnitude but opp direc^n)

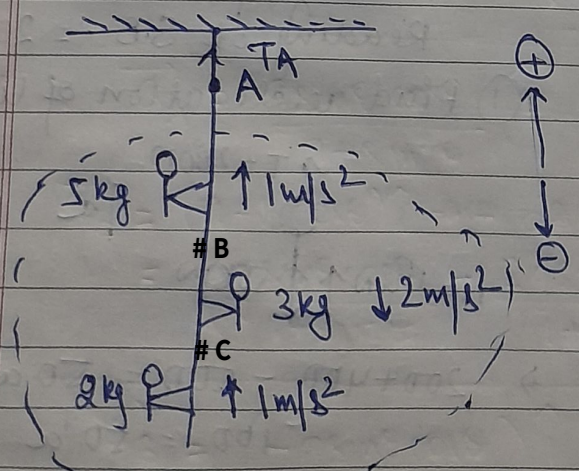
$\therefore \boxed{\vec{F_{ext}} = m_1 \vec{a_1} + m_2 \vec{a_2} + \dots + m_n \vec{a_n}}$

$F_{ext} = m_1 a_{1x} + m_2 a_{2x} + m_3 a_{3x} + \dots$

$F_{ext_H} = m_1 a_{1H} + m_2 a_{2H} + m_3 a_{3H} + \dots$

$F_{ext_V} = m_1 a_{1V} + m_2 a_{2V} + m_3 a_{3V} + \dots$

Q

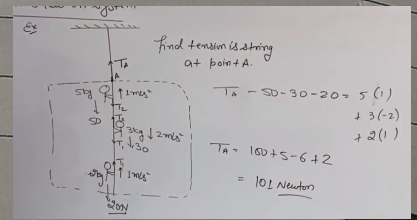


Find tension in string at point A.

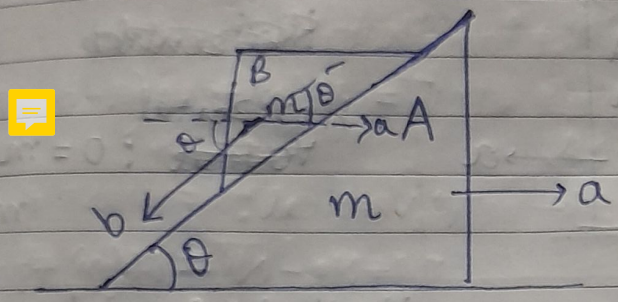
$T_A - 50 - 30 - 20 = 5(1) + 3(-2) + 2(1)$

$T_A = 100 + 5 - 6 + 2$

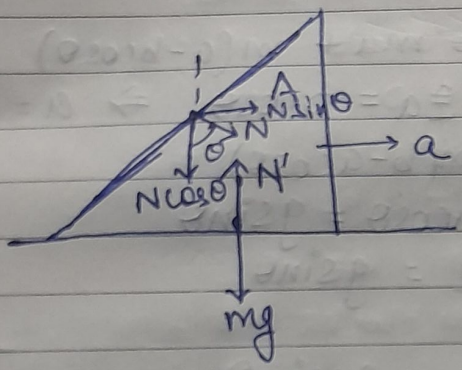
$\boxed{T_A = 101 N}$

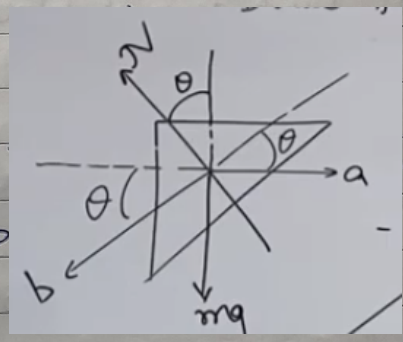


$$\vec{a}_{BA} = \vec{a}_B + \vec{a}_A$$
$$\vec{a}_B = \vec{a}_{BA} + \vec{a}_A$$



① Draw FBD of Block A and Block B

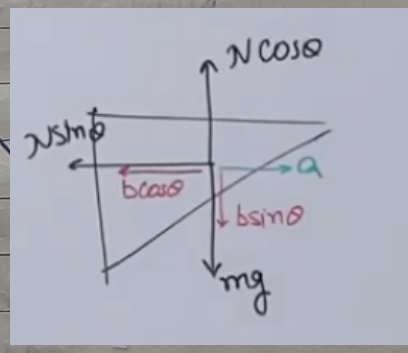




② Apply newtons law in Horizontal and Vertical direction:

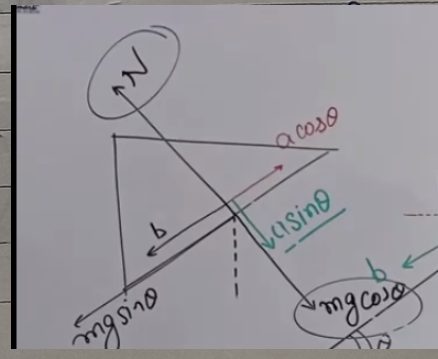
a) Block A : Hori : $N\sin\theta = ma$ — (1)
Vert : $N' = N\cos\theta + mg$ — (2)

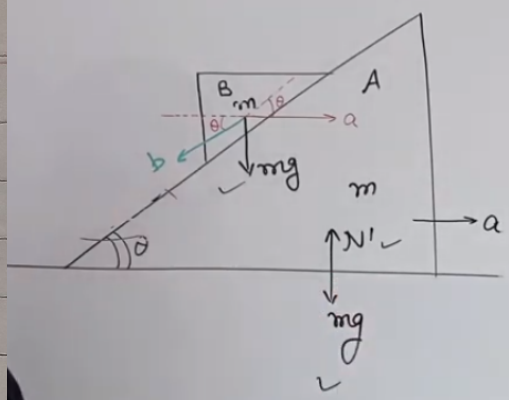
b) Block B:



Hori : $N\sin\theta = m(b\cos\theta - a)$ — (3)
Vert : $mg - N\cos\theta = mb\sin\theta$ — (4)
Along IP : $mg\sin\theta = m(b - a\cos\theta)$ — (5)
⊥ to IP : $mg\cos\theta - N = ma\sin\theta$ — (6)

③ Apply Newtons Law on (A+B) as system in Horizontal and vertical.





Draw fbd of (A+B)

apply NL on A+B

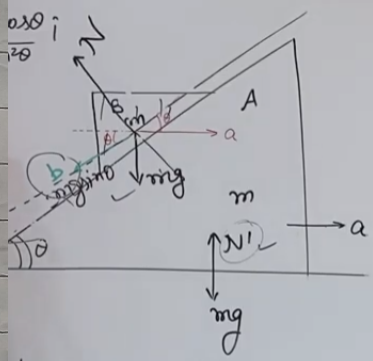
in

Vert: $2mg - N' = m(b\sin\theta) + m(0)$ — (7)

Hor²: $0 = m(a - b\cos\theta) + m(a)$ — (8)

$m(b\sin\theta)$ — (8)

(4) Find accn of blocks:



Find accn of blocks.

NL on (A+B) along

Hor²: $0 = ma + m(a - b\cos\theta)$ — (1)

$2a = b\cos\theta$ — (1)

On B, along I.P

$= mg\sin\theta = m(b - a\cos\theta)$

$b - a\cos\theta = g\sin\theta$ — (2)

$b - \frac{b\cos\theta}{2}\cos\theta = g\sin\theta$

$b\left\{1 - \frac{\cos^2\theta}{2}\right\} = g\sin\theta \Rightarrow \frac{b(2-\cos^2\theta)}{2} = g\sin\theta$

$a = \frac{g\sin\theta\cos\theta}{1+\sin^2\theta}$

$b = \frac{2g\sin\theta}{2-\cos^2\theta} = \frac{2g\sin\theta}{2-(1-\sin^2\theta)} = \frac{2g\sin\theta}{1+\sin^2\theta}$

competishu

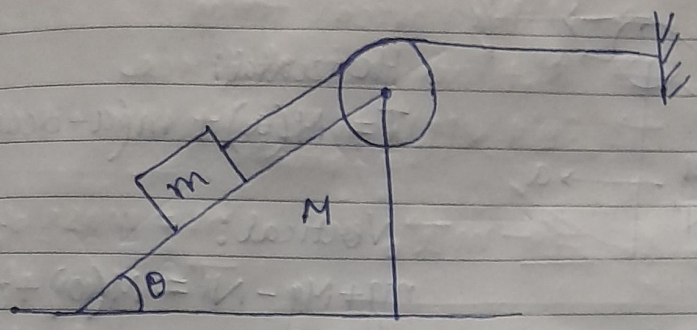
$\therefore \boxed{\vec{a}_A = \frac{g\sin\theta\cos\theta}{1+\sin^2\theta}\hat{i}}$

$\vec{a}_B = (a - b\cos\theta)\hat{i} - b\sin\theta\,\hat{j}$

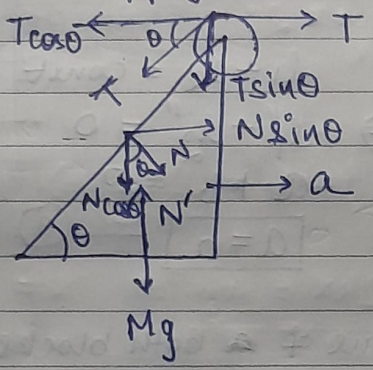
$= \left(\frac{g\sin\theta\cos\theta}{1+\sin^2\theta} - \frac{2g\sin\theta\cos\theta}{1+\sin^2\theta}\right)\hat{i} - \frac{2g\sin^2\theta}{1+\sin^2\theta}\hat{j}$

$\boxed{\vec{a}_B = \left(\frac{-g\sin\theta\cos\theta}{1+\sin^2\theta}\right)\hat{i} - \left(\frac{2g\sin^2\theta}{1+\sin^2\theta}\right)\hat{j}}$

Q



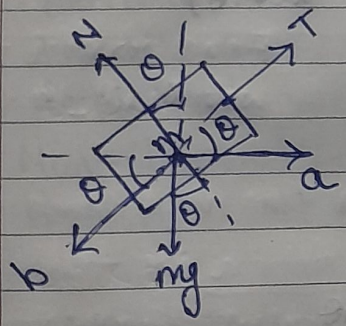
① Draw FBD of M and apply Newtons Law in Hor^2 and Verti:

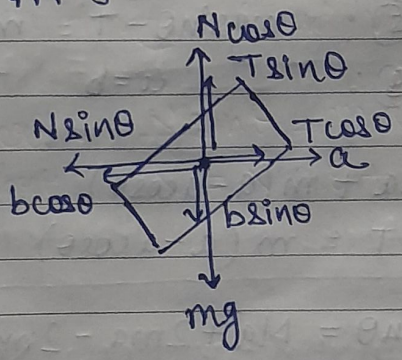


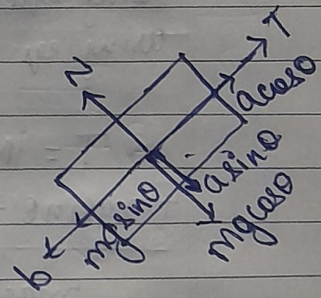
Hori: $T + N\sin\theta - T\cos\theta = Ma$ — ②

Vert: $Mg + N\cos\theta = N' + T\sin\theta$ — ②

② Draw FBD of m and apply NL:







Hori: $T\cos\theta - N\sin\theta = m(a - b\cos\theta)$ — ③

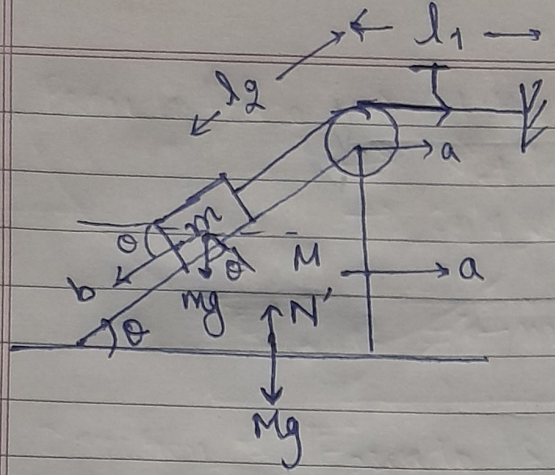
Vert: $mg - N\cos\theta - T\sin\theta = mb\sin\theta$ — ④

Along IP: $mg\sin\theta - T = m(b - a\cos\theta)$ — ⑤

⊥ to IP: $mg\cos\theta - N = ma\sin\theta$ — ⑥

③ Draw FBD of (M+m) and NL on this system along Horizontal and Vertical.

Page No.
Date:



Horizontal:

$$T = M(a) + m(a - b\cos\theta) \quad -(7)$$

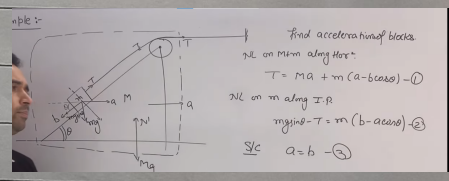
Vertical:

$$mg + Mg - N' = M(0) + m(b\sin\theta)$$

(4) Apply string constraint and relate a and b:

$$l_1 + l_2 = \text{const}$$
$$\Rightarrow l_1'' + l_2'' = 0$$
$$= -a + b = 0$$
$$= \boxed{a = b} \quad -(9)$$

(5) Find accelerations of both blocks



Using eqn (7): $T = Ma + m(a - b\cos\theta)$

Using eqn (5): $mg\sin\theta - T = m(b - a\cos\theta)$

Using eqn (9): $a = b$

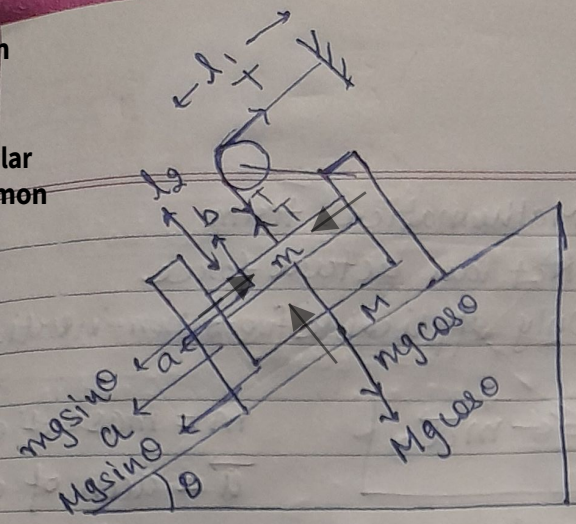
$$T = Ma + m(a - a\cos\theta)$$
$$+\ mg\sin\theta - T = m(a - a\cos\theta)$$

$$mg\sin\theta = Ma + 2ma - 2ma\cos\theta$$

$$\boxed{a = \frac{mg\sin\theta}{M + 2m - 2m\cos\theta} = b}$$

$$\vec{a_m} = (a - b\cos\theta)\hat{i} - b\sin\theta\,\hat{j}$$

b is relative acceleration of m wrt M, which is perpendicular to the common surface



Find acc^n of blocks

① M+m along I.P. :

$mg\sin\theta + Mg\sin\theta - T = Ma + ma$ — ①

② NL on 'm' ⊥ to IP :

$T - mg\cos\theta = mb = ma$ — ②

Adding ① and ②

$mg\sin\theta + Mg\sin\theta - mg\cos\theta = (M+2m)a$

$$a = \frac{mg\sin\theta + Mg\sin\theta - mg\cos\theta}{M+2m}$$

③ String constant:

$l_1 + l_2 = \text{const}$

$\Rightarrow l_1'' + l_2'' = 0$

$\Rightarrow a + (-b) = 0$

$a = b$ — ③

$$\vec{a}_M = a$$
$$\vec{a}_m = \sqrt{2}\,a$$

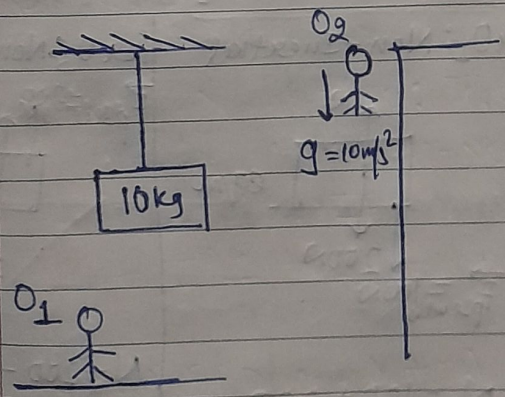
pseudo = fake/ false

Pseudo Force :

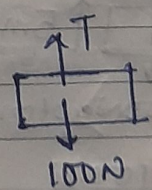
→ Inertial frame of reference : It is non-accelerating ref. frame ($\vec{v}$ = const or zero)
↳ NLM are valid

→ Non-Inertial frame of reference : It is accelerating ref. frame (variable velocity)
↳ NLM are not valid

Ex

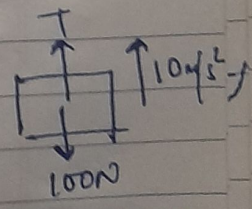


FBD Acc to O_1



$T = 100N$ (Correct)

FBD Acc to O_2



$T - 100 = 10 \times 10$

$T = 200N$ (Wrong)

Different values of tension are not possible

∴ We need pseudo force as NLM is not valid for Observer 2

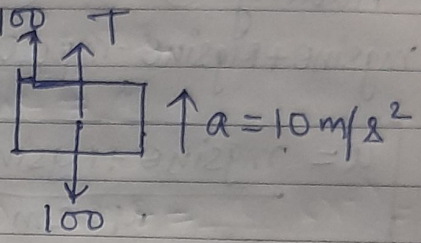
Pseudo force → Mathematical term
→ Not an actual force
→ Only applicable from non-inertial ref.

$$\boxed{\vec{F}_{pseudo} = -m\vec{a}_o}$$

m = mass of object
$\vec{a}$ = accn of observer

Direction of $\vec{F}_{pseudo}$ is opposite to accn of observer.

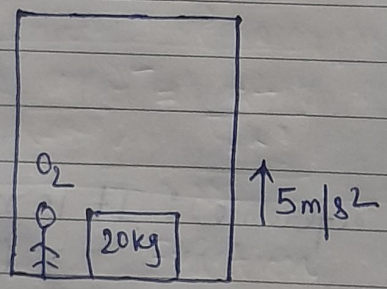
Now for observer 2: $F_{pseudo}=100$

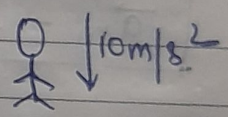


$T+100-100 = 10\times10$

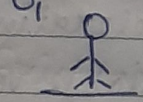
$\boxed{T = 100N}$

Ex

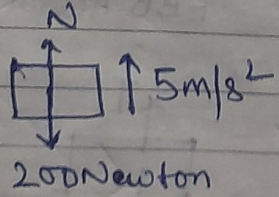




Find normal on block wrt O_1, O_2 and O_3



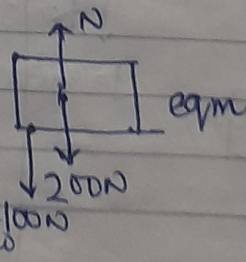
For O_1: Inertial frame



$N-200 = 100$

$N = 300$ Newton

For O_2: Non-Inertial



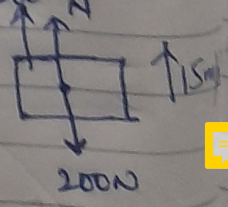
$\vec{F}_{pseudo} = 100N$

$N = 100+200$

$N = 300$ Newton

For O_3: Non-Inertial

$\vec{F}_{pseu} = 200$

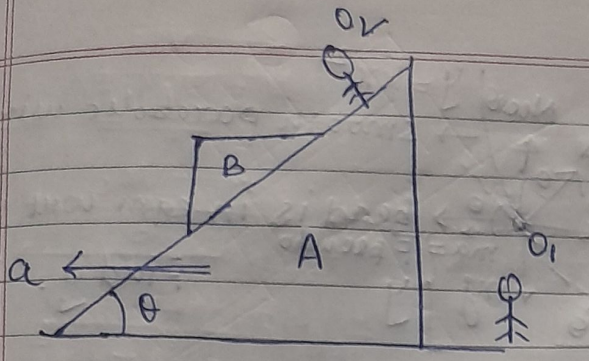


$N+200-200 = 20\times$

$N = 300$ Newton

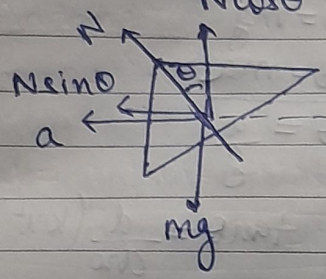
= 5+10

Q



Find 'a' such that block B doesnt slip over block A.

wrt O_1: Inertial



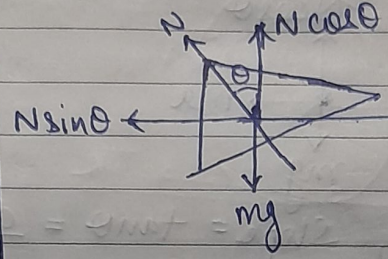
$N\cos\theta = mg$

$N\sin\theta = ma$

$\tan\theta = \frac{a}{g}$

$\boxed{a = g\tan\theta}$

wrt O_2: Non-Inertial



eqm

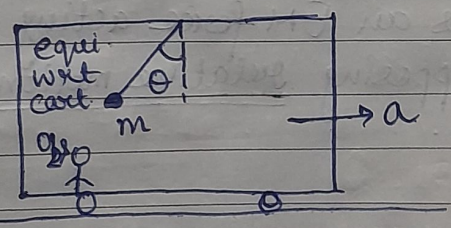
$N\cos\theta = mg$

$N\sin\theta = ma$

$\tan\theta = \frac{a}{g}$

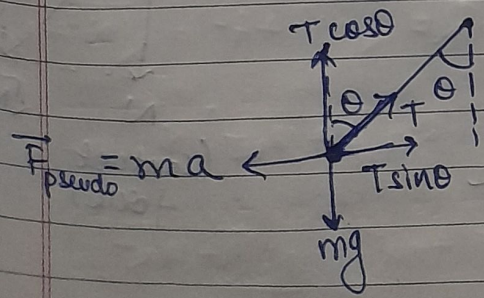
$\boxed{a = g\tan\theta}$

Q



Find θ

Observer is in non-inertial frame

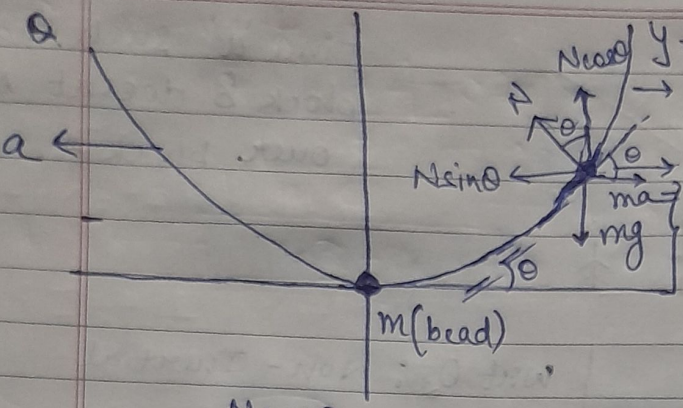


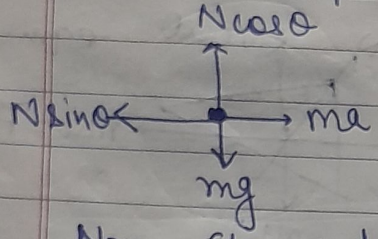
$T\cos\theta = mg$

$T\sin\theta = ma$

$\tan\theta = a/g$

$\boxed{\theta = \tan^{-1}\left(\frac{a}{g}\right)}$





$N\cos\theta = mg$

$N\sin\theta = ma$

$\tan\theta = \frac{a}{g}$

Now, Slope $= \tan\theta = \frac{dy}{dx} \Rightarrow \tan\theta = \frac{d}{dx}(kx^2) = 2kx$

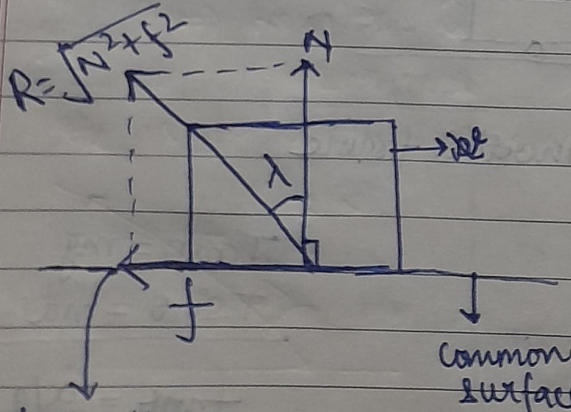
$\therefore \quad 2kx = \frac{a}{g} \Rightarrow \boxed{x = \frac{a}{2kg}}$

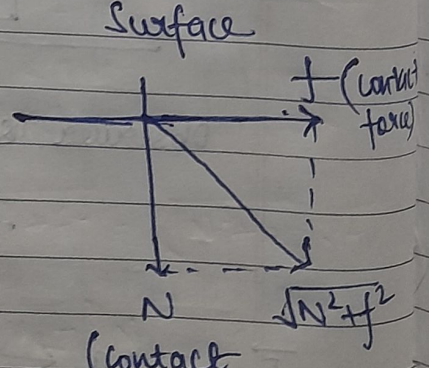
$$\boxed{y = kx^2 = k\left(\frac{a}{2kg}\right)^2 = \frac{a^2}{4kg^2}}$$

Friction

(by Abhinav: zig-zag surface means surface is rough)

Frictional force is an EM force acting along common rough surface, always opposing relative motion or tendency of relative motion.





Page No.
Date:

λ = Angle of Friction: $\tan\lambda = \frac{f}{N} \Rightarrow \boxed{\lambda = \tan^{-1}\left(\frac{f}{N}\right)}$

Types of Frictional force:

① Kinetic friction: It will act if there is relative motion between bodies.

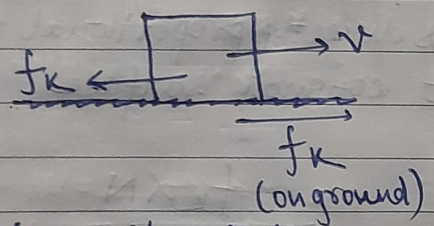
② Static friction: No relative motion. (Only tendency of relative motion)

Kinetic Friction:

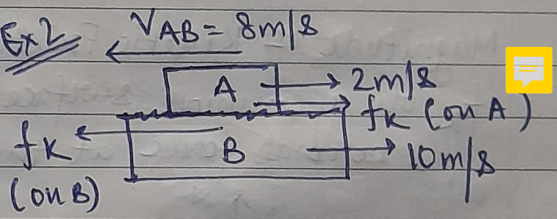
→ It opposes relative motion.

→ Direction: Always act opposite to relative velocity.

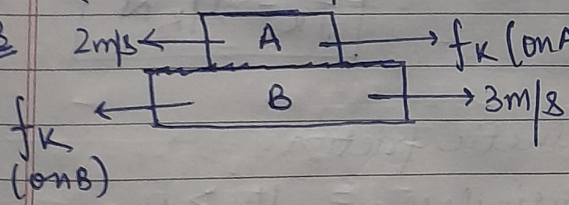
Ex1



Ex2

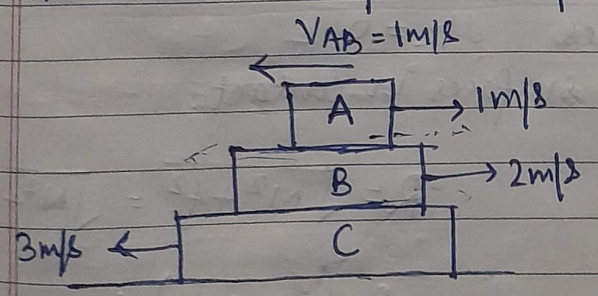


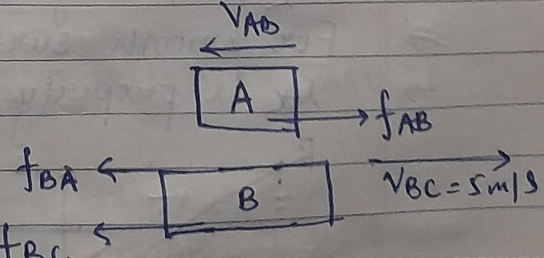
Ex3

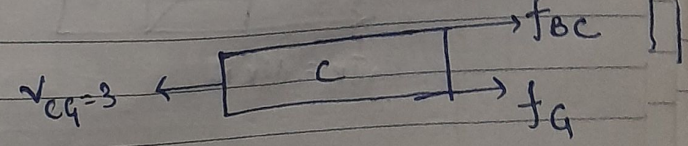


Q Find direction of Kinetic friction.

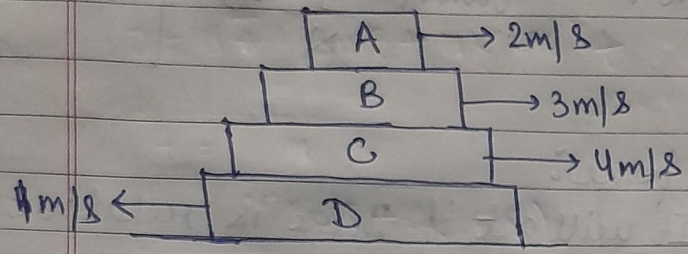
1)

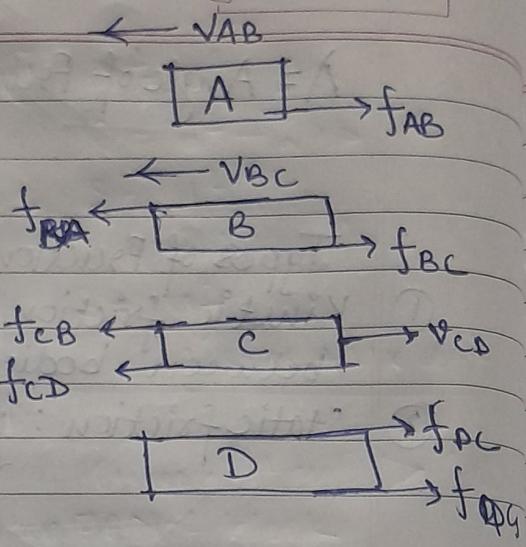




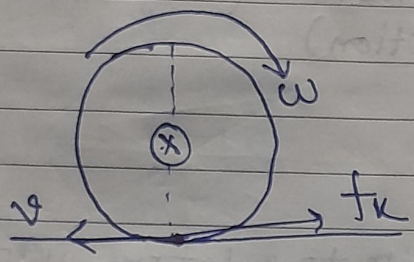


2)



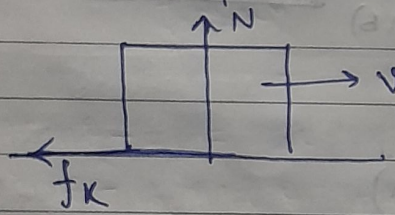


3)



Magnitude of Kinetic Friction:

f_K on a common surface is directly proportional to normal force at that common surface.



$$f_K \propto N$$

$$\boxed{f_K = \mu_K N}$$

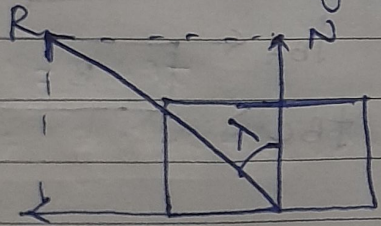
μ_K = coefficient of kinetic friction

↳ unitless, dimensionless, $\mu_K \geq 0$

→ $\mu_K \uparrow \Rightarrow$ Roughness ↕

→ For smooth surfaces $\mu_K = 0$

→ μ_K is property of surfaces in contact.

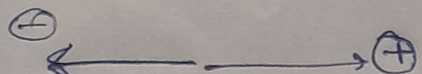


$$R = \sqrt{N^2 + f^2} = \sqrt{N^2 + \mu_K^2 N^2}$$

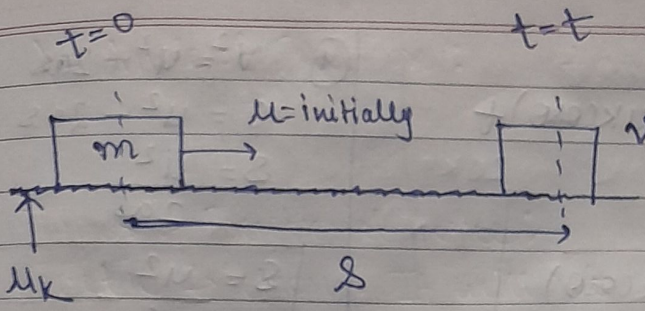
$$R = \left(\sqrt{1 + \mu_K^2}\right) N$$

$$\lambda = \tan^{-1}\left(\frac{N\left(\sqrt{1+\mu_K^2}\right)}{N}\right)$$

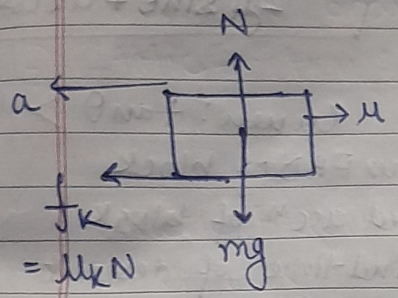
Page No.
Date:



Q



① Find FBD of block
② Apply NL and find accn of block
③ Find the time when block stops.
④ Find the distance travelled by block before it stops.



$N = mg$

$f_k = \mu_k N = \mu_k mg$

$f_k = ma$

$\mu_k mg = ma$

$\boxed{a = \mu_k g}$

③ $v = u + at$

$0 = u + (\mu_k g)t$

$\boxed{t = \frac{u}{\mu_k g}}$

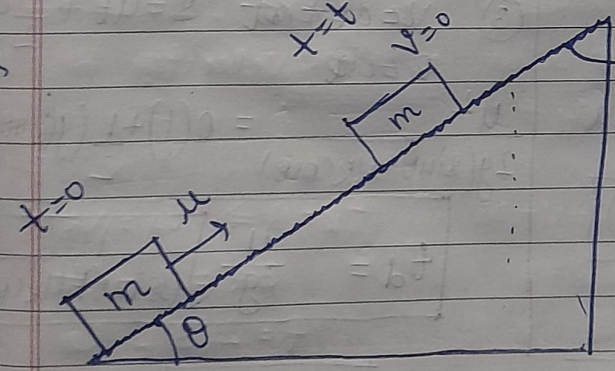
④ $v^2 - u^2 = 2as$

$0 - u^2 = 2(-a)s$

$s = \frac{u^2}{2a}$

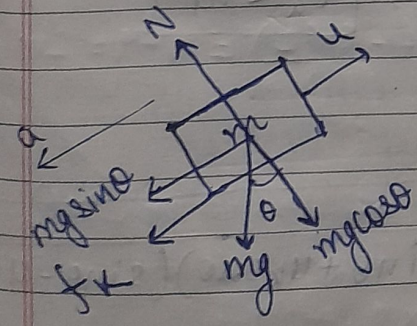
$\boxed{s = \frac{u^2}{2\mu_k g}}$

Q



Given: $\tan\theta > \mu_k$

① Draw FBD of block
② Apply NL and find accn
③ Find time of ascent
④ Find distance travelled while ascending.



$f_k = \mu_k N = \mu_k mg\cos\theta$

$f_k + mg\sin\theta = ma$

$\mu_k mg\cos\theta + mg\sin\theta = ma$

$\boxed{a = g(\mu_k\cos\theta + \sin\theta)}$

③ $v = u + at$

$0 = u - g(\sin\theta + \mu_k \cos\theta)\, t$

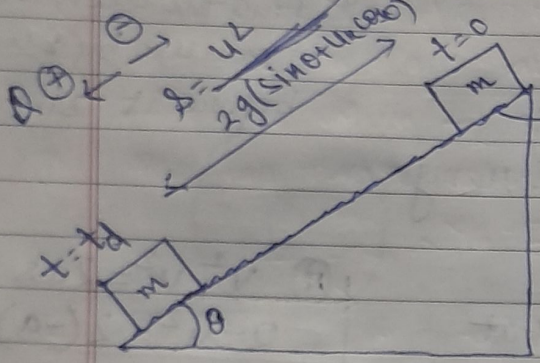
$$t_A = \frac{u}{g(\sin\theta + \mu_k \cos\theta)}$$

④ $v^2 = u^2 + 2as$

$0 = u^2 - 2as$

$s = \frac{u^2}{2a}$

$$s = \frac{u^2}{2g(\sin\theta + \mu_k \cos\theta)}$$



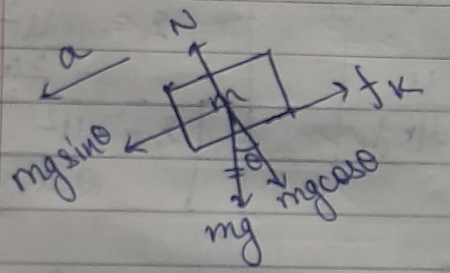
Given: $\tan\theta > \mu_k$

① Draw FBD of block

② Find acc^n of block

③ Find time of descent

④ Find speed of block when it reaches the base



$N = mg\cos\theta$

$f_k = \mu_k N = \mu_k mg\cos\theta$

$mg\sin\theta - f_k = ma$

$mg\sin\theta - \mu_k mg\cos\theta = ma$

$$a = g(\sin\theta - \mu_k \cos\theta)$$

$\mu_k \downarrow$ a ↑

③ $s = ut + \frac{1}{2}at^2$

$$\frac{u^2}{2g(\sin\theta + \mu_k\cos\theta)} = 0(t) + \frac{1}{2}(g(\sin\theta - \mu_k\cos\theta))\, t^2$$

$$t_d = \frac{u}{g}\sqrt{\frac{1}{(\sin\theta + \mu_k\cos\theta)(\sin\theta - \mu_k\cos\theta)}}$$

$\frac{t_a}{t_d} = \sqrt{\frac{\sin\theta - \mu_k\cos\theta}{\sin\theta + \mu_k\cos\theta}} < 1 \Rightarrow t_a < t_d$

④ $v = u + at$

$$v = g(\sin\theta - \mu_k\cos\theta)\left(\frac{u}{g}\sqrt{\frac{1}{(\sin\theta + \mu_k\cos\theta)(\sin\theta - \mu_k\cos\theta)}}\right)$$

$$v = u\sqrt{\frac{\sin\theta - \mu_k\cos\theta}{\sin\theta + \mu_k\cos\theta}}$$

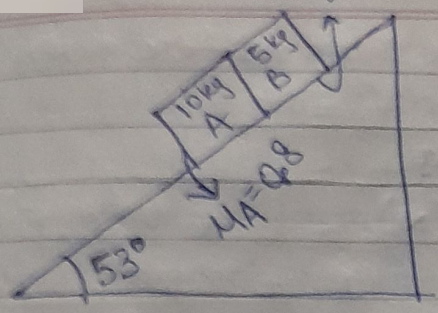
$v < u$

$a = g(\sin\theta - \mu\cos\theta)$

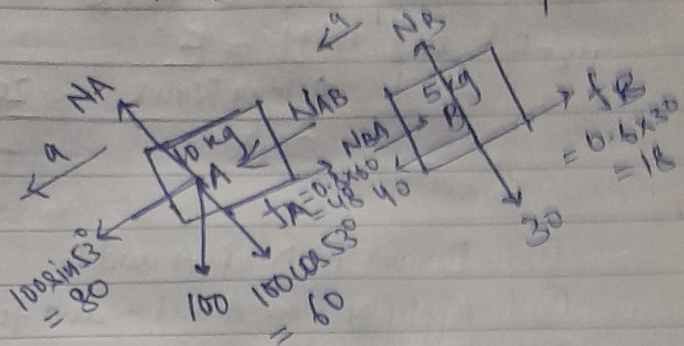
$\mu \downarrow \; a \uparrow$

Page No.
Date:

Q



Find accelerations of blocks



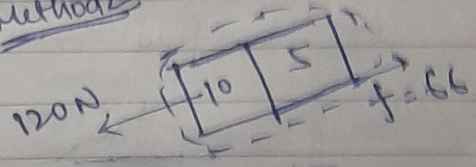
$80 + N - 48 = 10a$

$+ \; 40 + N - 18 = 5a$

$54 = 15a$

$a = \frac{54}{15} = 3.6 \; m/s^2$

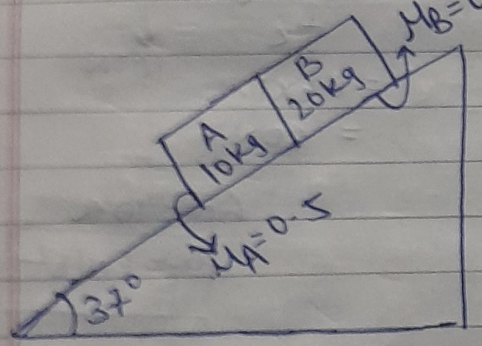
Method 2



$120 - 66 = 15a$

$a = \frac{54}{15} = 3.6 \; m/s^2$

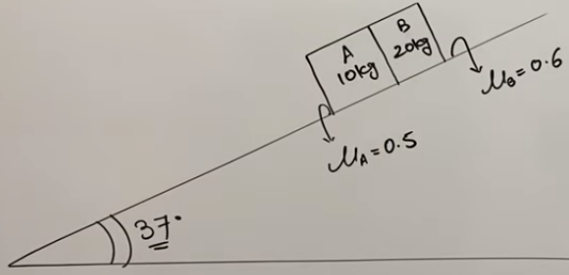
Q

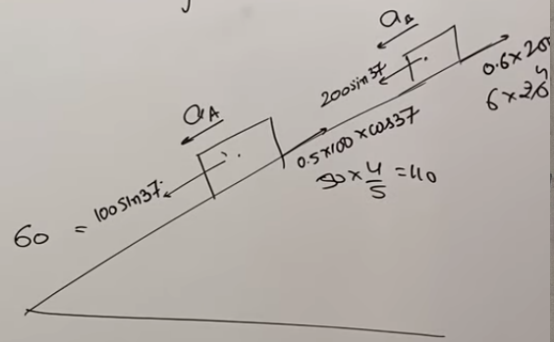


Find accelerations of blocks

Ex

Find accelerations of blocks.





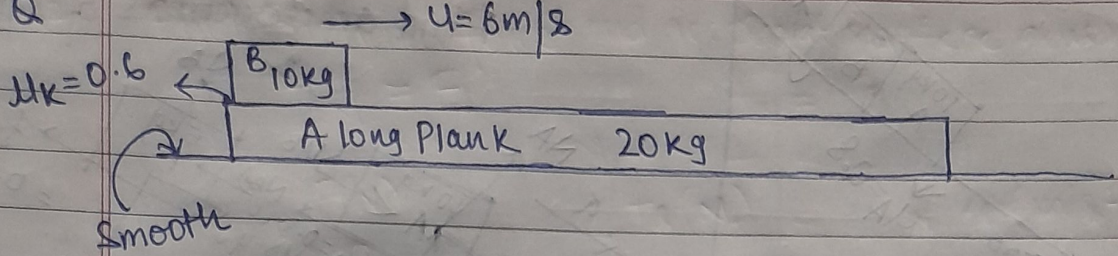
$20 = 10 a_A$

$a_A = 2 \; m/s^2$

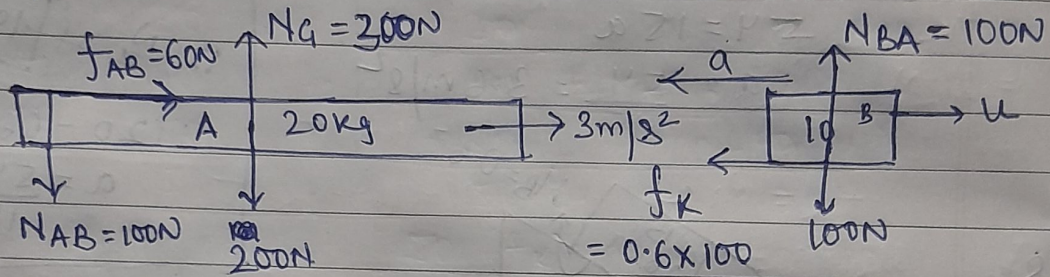
$120 - 96 = 20 a_B$

$\frac{24}{20} = \frac{6}{5} = 1.2 = a_B$

Q



① Draw FBD of A and B
② Apply NL on A and B and find acc^n
③ Find time when f_k stops acting
④ Also find their velocities at this instant



$f_k = 0.6 \times 100 = 60N$

$60 = 20 \times a$

$\boxed{a_A = 3m/s^2}$

$60 = 10a$

$\boxed{a_B = 6m/s^2}$

③ Concept: f_k will stop acting when relative velocity b/w A and B becomes $0 \Rightarrow V_A = V_B$

$V_B = 6 - 6t \quad (v = u + at)$

$V_A = 0 + 3t$

$f_k = 0 \Rightarrow V_A = V_B$

$3t = 6 - 6t$

$\boxed{t = 2/3 \text{ sec}}$

④ $V_A = 3\left(\frac{2}{3}\right) = 2m/s \qquad V_B = 6 - 6\left(\frac{2}{3}\right) = 6 - 4 = 2m/s$

Alternate:

Momentum conservation (To be studied in COM):

$P_i = P_f$

$10 \times 6 + 20 \times 0 = 10 \times v + 20v$

$60 = 30v$

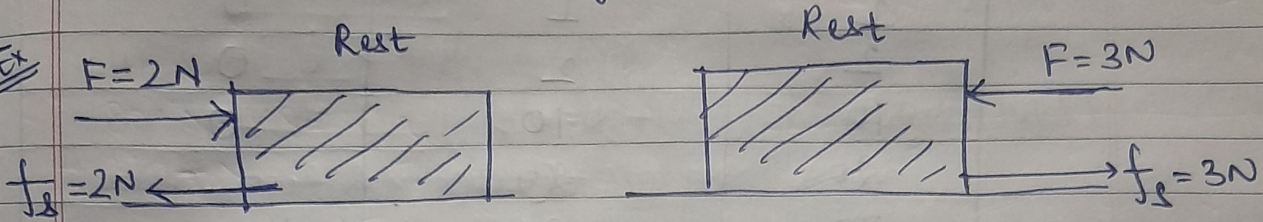
$\boxed{v = 2m/s}$

Page No.
Date:

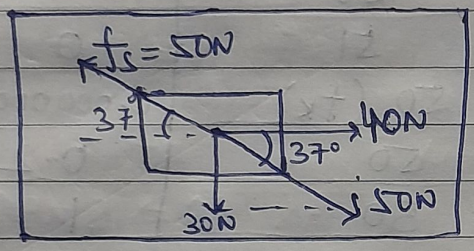
Static Friction:

→ There will be no relative motion b/w the objects in motion
→ There will be tendency of relative motion.
→ It is a self-adjusting force (Magnitude and direction of static friction may vary to avoid any relative motion).

Ex



Q TOP VIEW



Find friction acting on the block.

Limiting Friction: Maximum possible value of static friction

$$f_{smax} = f_{lim}$$

$$0 \le f_s \le f_{lim}$$

Exact value of f_s is decided by situation.

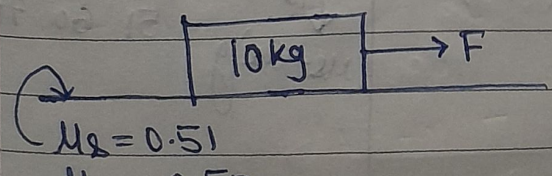
$$f_{smax} = f_{lim} = \mu_s N$$

μ_s ⇒ Coefficient of static friction
→ Property of bodies in contact
$\mu_s \propto$ Roughness

unitless & dimensionless

Practically: $\mu_s \geq \mu_k$

Q



① Find limiting friction

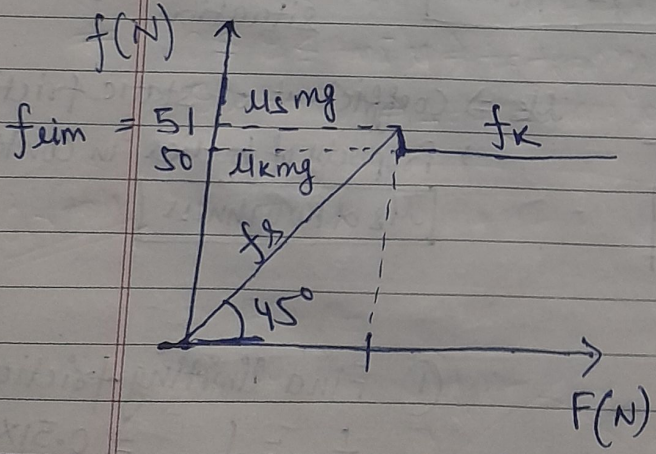
$f_{lim} = f_{smax} = 0.51 \times 100 = 51N$

Page No.
Date:

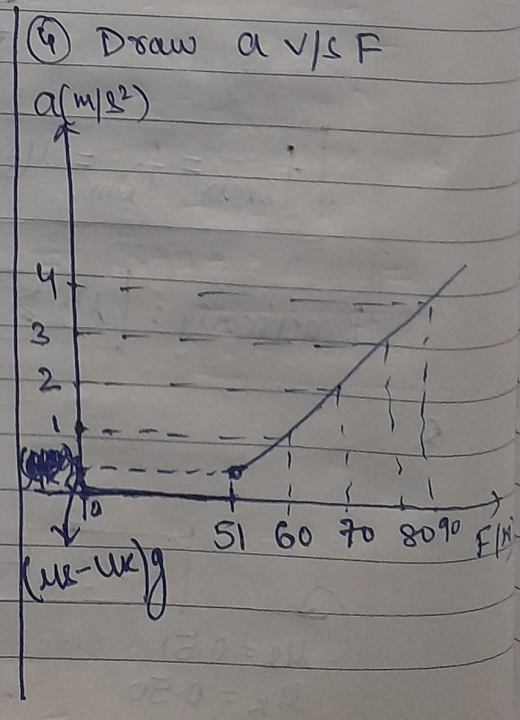
② Find frictional force and acceleration for given values of $f_{ext} = F$

F(N)	f_s(N)	a (m/s²)
0	0	0
1	1	0
2	2	0
10	10	0
20	20	0
30	30	0
40	40	0
50	50	0
51	51	0
51.00000001	50 (f_k)	0.100000001
52	50	0.2
60	50	1
70	50	2
80	50	3
100	50	5

③ Draw f v/s F



④ Draw a v/s F



Page No.
Date:

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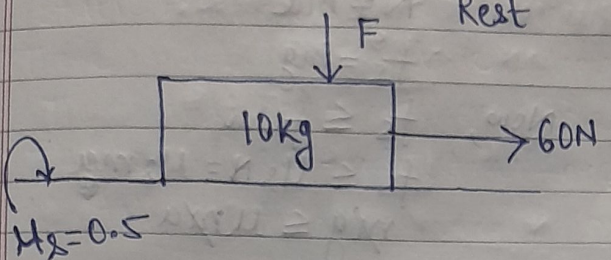
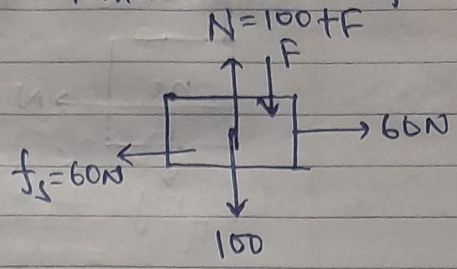


Rest

Find the value of F.



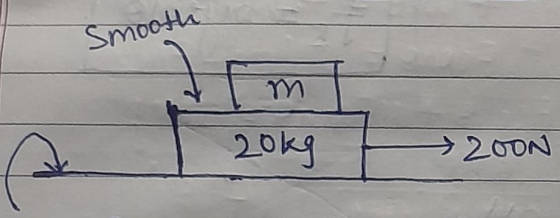
We know $f_s \leq f_{lim}$

$f_s \leq \mu_s N \Rightarrow 60 \leq 0.5(100+F)$

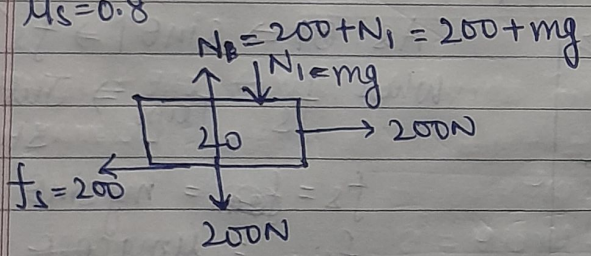
$120 \leq 100 + F$

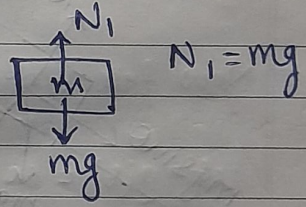
$\boxed{F \geq 20N}$

Q



Find the value of m such that 20kg remains at rest.





$f_s \leq f_{lim}$

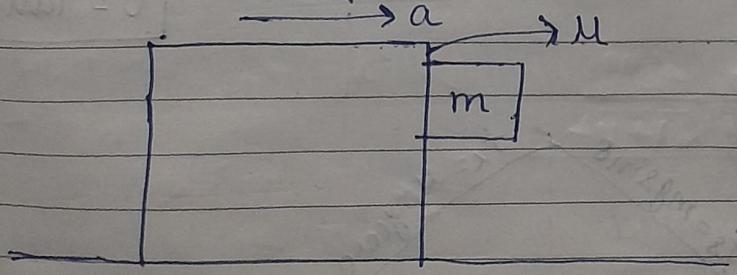
$= 200 \leq 0.8(200 + mg)$

$= 200 \leq 80(20 + m)$

$= \frac{200}{8} \leq (20+m) \Rightarrow m \geq \frac{200}{8} - 20$

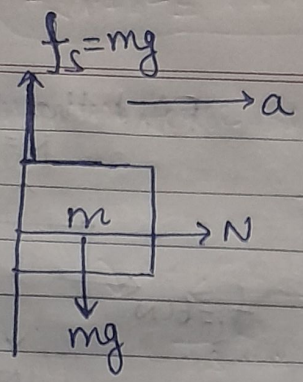
$\boxed{m \geq 5kg}$

Q



Find the value of 'a' such that m doesn't fall.

Neelgagan Page No. Date:



$N = ma$

$f_s = mg$

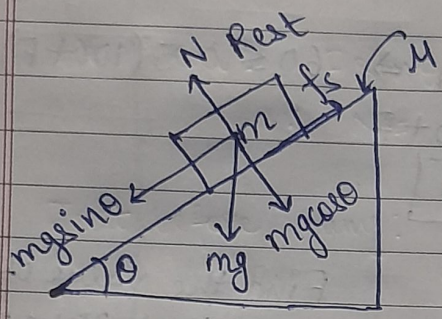
Now, $f_s \le f_{lim}$

$f_s \le \mu_s N = \mu_s ma$

$mg \le \mu ma$

$$\boxed{a \ge \frac{g}{\mu}}$$

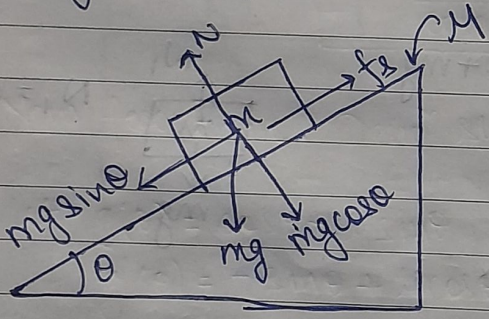
Q



Find frictional force acting on the block.

$$\boxed{f_s = mg\sin\theta}$$

Angle of Repose :



($\theta\uparrow$ (slowly))

As $\theta\uparrow \Rightarrow \sin\theta\uparrow \Rightarrow f_s\uparrow$

When $f_s = f_{lim} \Rightarrow$ Just slipping.

$f_s = f_{lim} = mg\sin\theta$

$f_{lim} = \mu mg\cos\theta$

For just slipping.

$f_s = f_{lim}$

$mg\sin\theta = \mu mg\cos\theta$

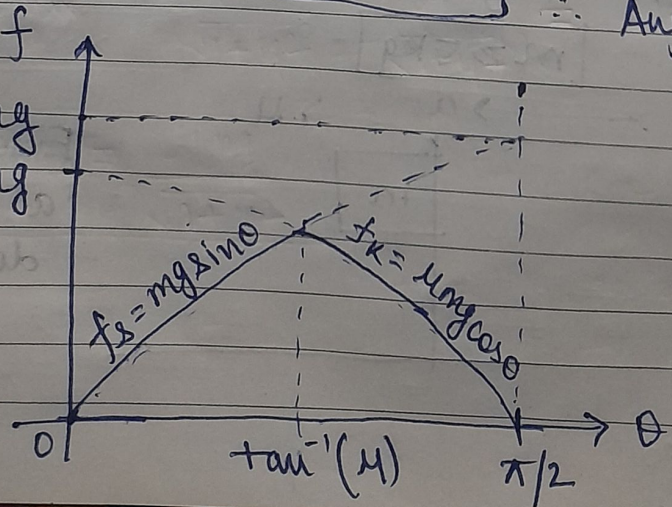
$\tan\theta = \mu$

$\therefore$ Angle of repose =

$$\boxed{\theta = \tan^{-1}\mu}$$

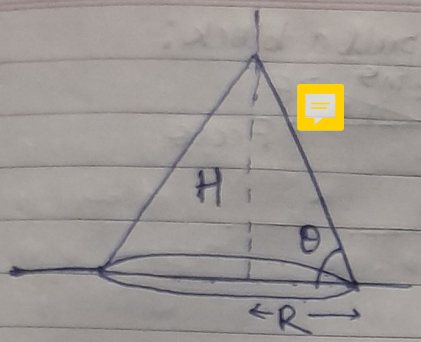
\# $\boxed{\begin{array}{l} 0 \le \theta \le \tan^{-1}\mu \Rightarrow f = f_s = mg\sin\theta \\ \theta > \tan^{-1}\mu \Rightarrow f = f_k = \mu mg\cos\theta \end{array}}$

Graph



assuming $(\mu < 1)$

Q



Coeff of friction b/w sand particles $= \mu$

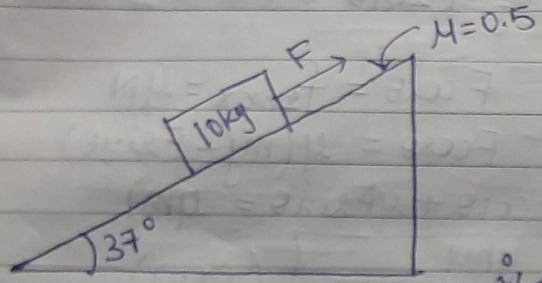
What can be max^m volume of this cone?

$V = \frac{1}{3}\pi R^2 H$ $\quad V\uparrow \Rightarrow H\uparrow$

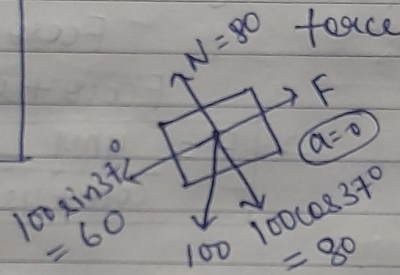
$\tan\theta = \frac{H}{R} = \mu \Rightarrow \boxed{H = \mu R}$

$\therefore \boxed{V = \frac{1}{3}\pi \mu R^3}$

Q



① Find the value of F so that there is no frictional force.

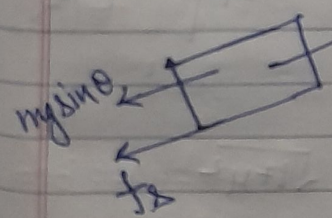


$\boxed{F = 60N}$

② What can be f_{lim} on block?

$f_{lim} = \mu N = 0.5 \times 80 = 40N$

③ Find min^m force required up the incline to just move the block up the incline.

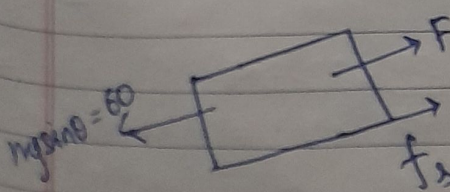


$\boxed{F > mg\sin\theta + f_{s\max}}$

$F > 60 + 40 \Rightarrow \boxed{F > 100N}$

$\therefore$ 100N is max^m force up the incline so that block doesn't move.

④ Find min^m force up the incline so that block doesn't move



$F_{min} + f_{lim} = mg\sin\theta$

$F_{min} = 60 - 40$

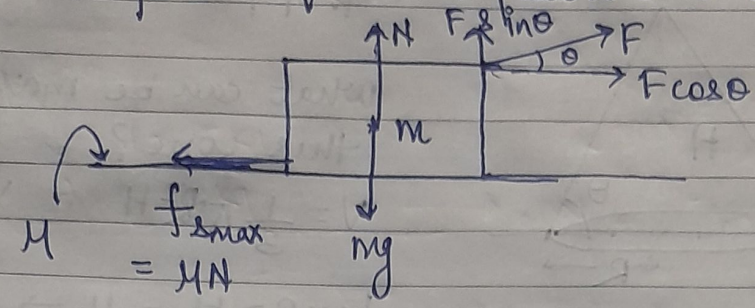
$\boxed{F_{min} = 20N}$

~~$a\sin\theta + b\sin\theta$~~

$$-\sqrt{a^2+b^2} \le a\sin\theta + b\cos\theta \le \sqrt{a^2+b^2}$$

Page No.
Date:

Minimum force required to pull a block:



Vertical: $F\sin\theta + N = mg$

$\therefore N = mg - F\sin\theta$

Just before moving: $F\cos\theta = f_{s\max} = \mu N$

$F\cos\theta = \mu(mg - F\sin\theta)$

$F\cos\theta + \mu F\sin\theta = \mu mg$

$$\therefore \boxed{F = \frac{\mu mg}{\cos\theta + \mu\sin\theta}} \quad (f(\theta))$$

for some value of θ $F \Rightarrow$ least.

Now for F_{min}, Deno^r should be max $\Rightarrow \cos\theta + \mu\sin\theta$ should be max.

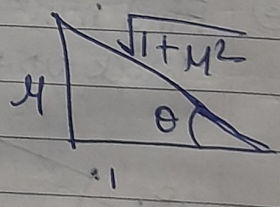
Let $g(\theta) = \cos\theta + \mu\sin\theta$

$g'(\theta) = -\sin\theta + \mu\cos\theta = 0$

$\Rightarrow \mu\cos\theta = \sin\theta$

$\Rightarrow \mu = \tan\theta$

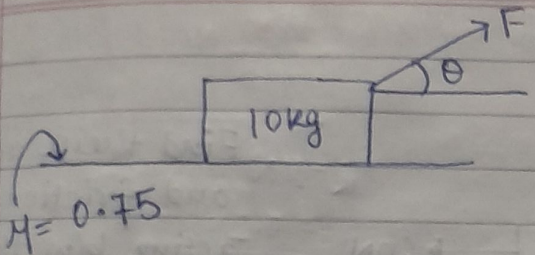
$$\Rightarrow \boxed{\theta = \tan^{-1}(\mu)}$$



$$g(\theta)_{max} = \frac{1}{\sqrt{1+\mu^2}} + \frac{\mu^2}{\sqrt{1+\mu^2}} = \sqrt{1+\mu^2}$$

$$\therefore \boxed{F_{min} = \frac{\mu mg}{\sqrt{1+\mu^2}}}$$

Q

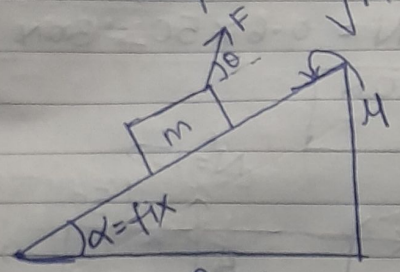


To pull this block with min^m force
① what should be θ ?
② what is min^m force?

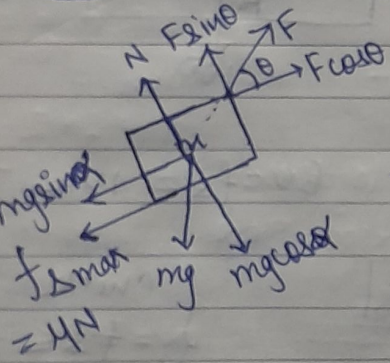
① $\theta = \tan^{-1}(\mu) = \tan^{-1}(0.75) = \tan^{-1}\left(\frac{3}{4}\right) = 37°$

② $F_{min} = \frac{\mu mg}{\sqrt{1+\mu^2}} = \frac{3/4 \times 10 \times 10}{\sqrt{1+\frac{9}{16}}} = 60N$

Q



Find the angle θ at which force required to pull the block up the incline is least. Also find min^m force.



$N + F\sin\theta = mg\cos\alpha$

$\therefore N = mg\cos\alpha - F\sin\theta$

Just before moving : $F\cos\theta = mg\sin\alpha + \mu N$

$F\cos\theta = mg\sin\alpha + \mu(mg\cos\alpha - F\sin\theta)$

$F\cos\theta + \mu F\sin\theta = mg\sin\alpha + \mu mg\cos\alpha$

$$F = \frac{mg\sin\alpha + \mu mg\cos\alpha}{\cos\theta + \mu\sin\theta} \quad f(\theta)$$

Max value of D^r $= \sqrt{1+\mu^2}$

$\therefore$ θ at which F is min $= \tan^{-1}(\mu)$

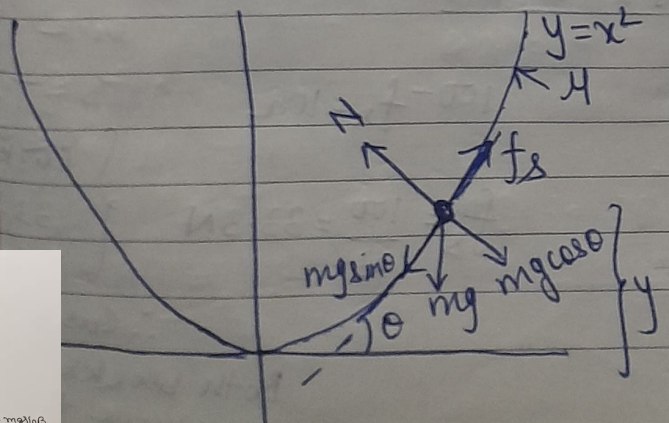
$$F_{min} = \frac{mg\sin\alpha + \mu mg\cos\alpha}{\sqrt{1+\mu^2}}$$

$g(\theta) = \cos\theta + \mu\sin\theta$

$g'(\theta) = -\sin\theta + \mu\cos\theta = 0$

$\tan\theta = \mu$

Q



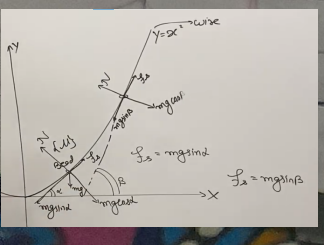
Find the max^m hight of bead so that it doesn't slip.

$f_{s max} = mg\sin\theta = \mu mg\cos\theta$

$\tan\theta = \mu$

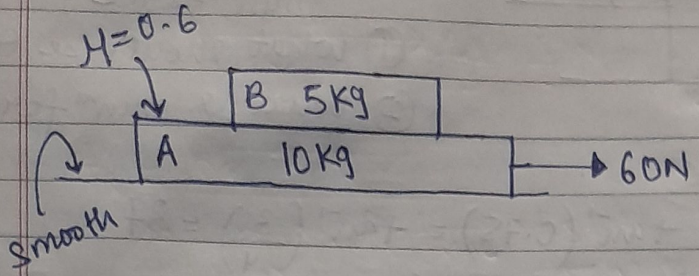
Also $\frac{dy}{dx} = 2x = \tan\theta = \mu$

$$x = \mu/2 \quad \therefore \quad y = x^2 = \mu^2/4$$



Two Block Problems :

Ex



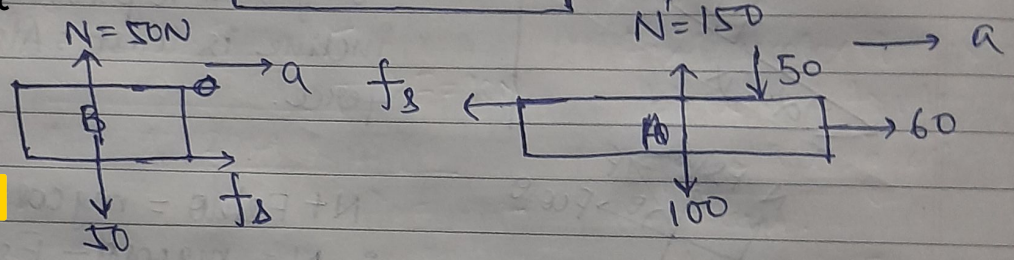
Find their accelerations and frictional force acting b/w them.

Case I : Let us assume both blocks are moving together :

since they are moving together, hence no relative motion and hence, friction is static

$$f_{lim} = \mu N = 0.6 \times 50 = 30N$$

$$\therefore \boxed{0 \le f_s \le 30N}$$



$f_s = 5a$ — (1)

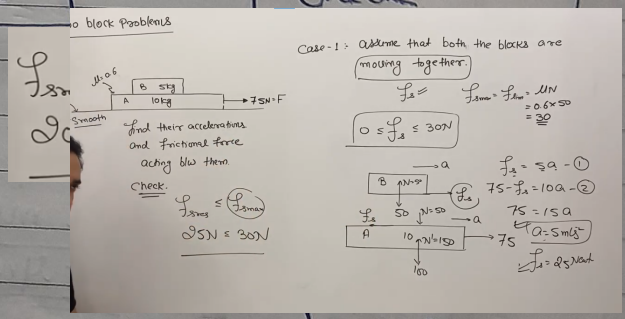
$60 - f_s = 10a$ — (2)

Add (1) and (2) $\quad 60 = 15a$

$\boxed{a = 4 m/s^2}$

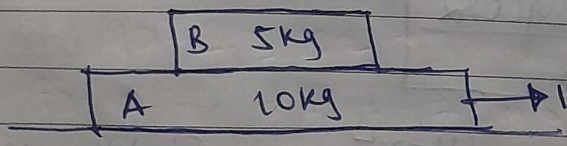
$\boxed{f_s = 5 \times 4 = 20N}$

(Check)



$f_{lim} = 30N$

Now if



$f_s = 5a$ — (1)

$100 - f_s = 10a$

Add (1) and (2)

$100 = 15a$

$a = \frac{20}{3} m/s^2$

$f_s = \frac{100}{3} = 33.3N$

Check

$33.3 \le 30$

False

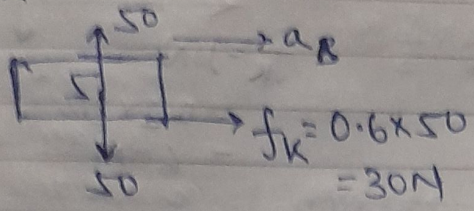
$\therefore$ Case I Failed

Both blocks will not move together.

Page No.
Date:

thus, the friction acting on the will be kinetic

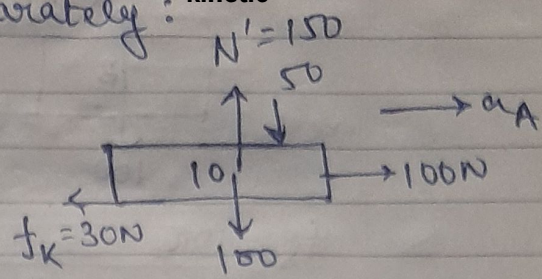
Both blocks are moving separately:
$\therefore f_k$ will act



$f_K = 5a_B$

$30 = 5a_B$

$\boxed{a_B = 6 m/s^2}$

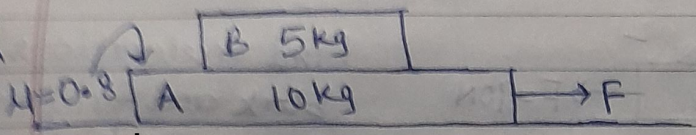


$100 - 30 = 10 a_A$

$70 = 10 a_A$

$\boxed{a_A = 7 m/s^2}$

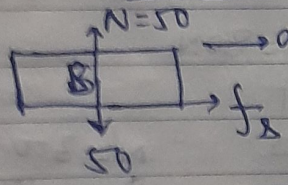
Q



Find max^m value of F so that there is no slipping b/w blocks.

$f_{lim} = \mu N = 0.8 \times 50 = 40N$

$\boxed{0 \le f_s \le 40N}$

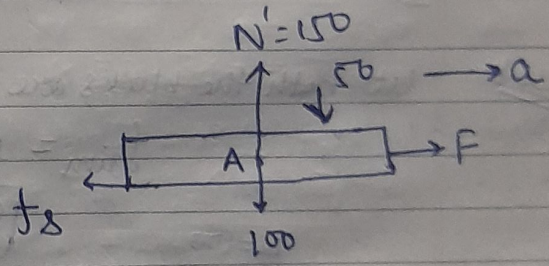


$f_s = 5a$ — (1)

For no slipping $f_s = f_{lim}$
add Fmax

$40 = 5a$

$\boxed{a = 8 m/s^2}$

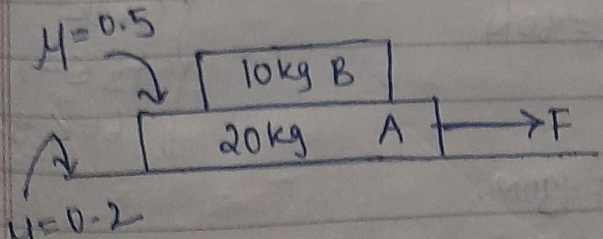


$F - f_s = 10a$

$F - 40 = 10 \times 8$

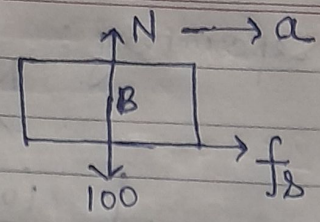
$\boxed{F = 120N}$

Q



Find max^m value of F so that there is no slipping b/w blocks.

Page No.
Date:



$f_s = 10a$

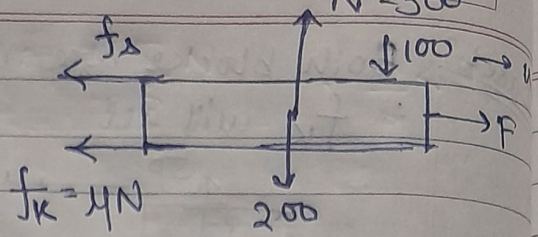
For no slipping, $f_s = f_{s\,lim}$ at f_{max}

$= 0.5 \times 100$

$= 50N$

$50 = 10a$

$\boxed{a = 5 m/s^2}$



$= 0.2 \times 300$

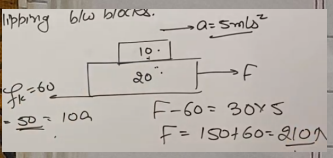
$= 60N$

$F - f_s - f_k = 20a$

$F - 50 - 60 = 20 \times 5$

$\boxed{F = 210N}$

slipping b/w blocks.

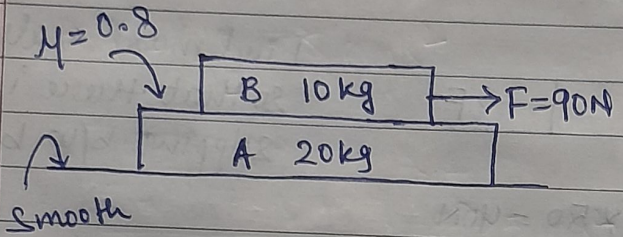


50 = 10a

F − 60 = 30 × 5

F = 150 + 60 = 210 N

Q

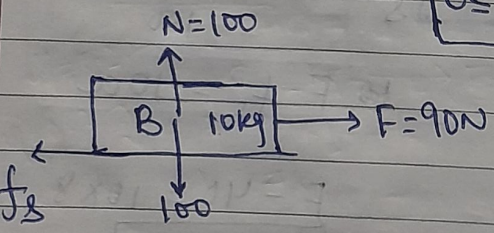


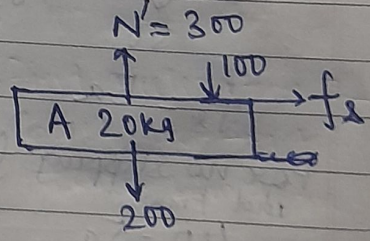
Find accelerations of block and frictional force acting between them.

Case I: Let both the blocks are moving together:

$f_{s\,lim} = \mu N = 0.8 \times 100 = 80N$

$\boxed{0 \le f_s \le 80}$





$90 - f_s = 10a$ — (1)

$f_s = 20a$ — (2)

From (1) and (2) $a = 3 m/s^2$

$f_s = 60N$

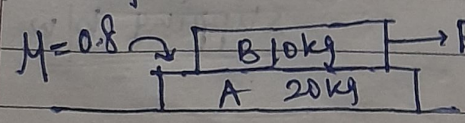
Check

$f_s \le f_{s\,lim}$

$60 \le 80$

∴ True

Q. Now if,



$f_{s\,lim} = \mu N = 0.8 \times 100$

$= 80N$

$130 - f_s = 10a$ — (1) $f_s = 20a$ — (2)

From (1) and (2)

$a = 4.33\ m/s^2$

$f_s = 86.66\ N$

check $f_s \le f_{lim}$

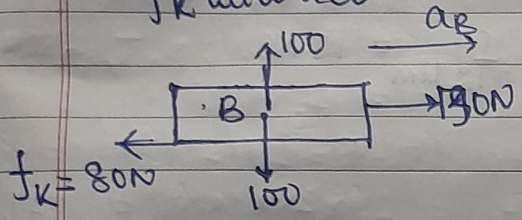
$86.66 \le 80$

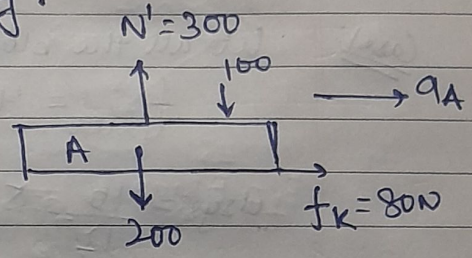
∴ case I fail

∴ Both block wont move together.

Case 2: Both blocks are moving separately:

∴ f_K will act





$130 - f_K = 10 a_B$

$130 - 80 = 10 a_B$

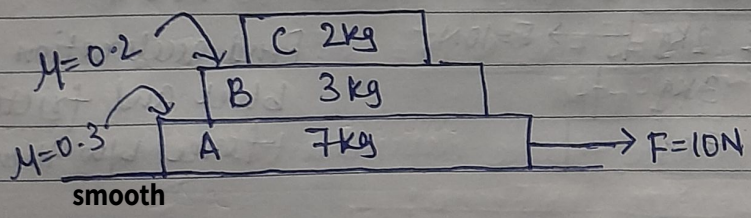
$\boxed{a_B = 5\ m/s^2}$

$80 = 20 a_A$

$\boxed{a_A = 4\ m/s^2}$

Three block Problem:

Ex



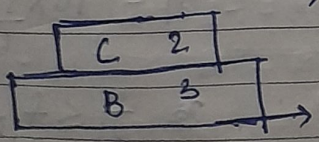
Find accelerations of blocks and frictional force between them.

Case I: Let all the blocks move together:

$f_{1lim} = 0.2 \times 20 = 4N$ $f_{2lim} = 0.3 \times 50 = 15N$

$a_{sys} = \frac{10N}{12kg} = \frac{5}{6}\ m/s^2$

$\rightarrow a = 5/6\ m/s^2$

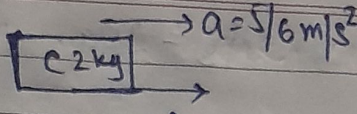


$f_2 = 5 \times \frac{5}{6} = \frac{25}{6}\ N$

$\therefore \boxed{f_2 = \frac{25}{6}}$

check $\frac{25}{6} \le 15$

∴ True



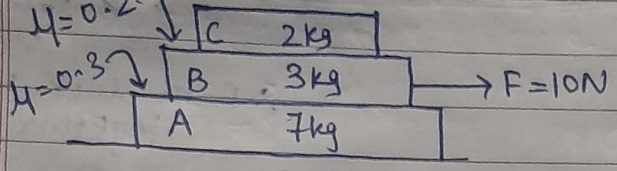
$f_{s_1} = 2 \times \frac{5}{6} = \frac{5}{3}$ N

$f_1 = \frac{5}{3}$ N

$\frac{5}{3} < 4$: True

case 2:

Q

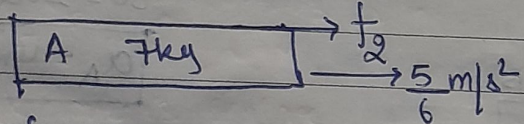


Find acceleration of block and frictional force b/w them

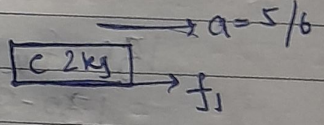
Case I: Let all the blocks move together:

$f_{1\,lim} = 4$N $\qquad f_{2\,lim} = 15$N

$a_{sys} = \frac{10}{12} = 5/6$ m/s²



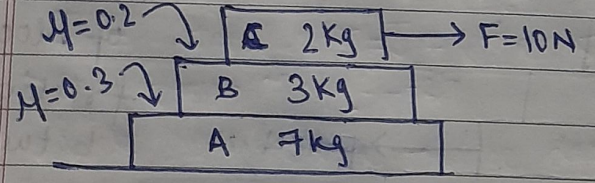
$f_2 = 7 \times \frac{5}{6} = \frac{35}{6}$ N



$f_1 = 2 \times \frac{5}{6} = \frac{5}{3}$ N

$\therefore$ $f_1 = \frac{5}{3}$ N $\qquad$ $f_2 = \frac{35}{6}$ N

Q

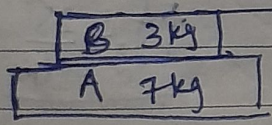


Find acceleration of block and frictional force b/w them.

Case I: Let all the blocks move together:

$f_{1\,lim} = 4$N $\qquad f_{2\,lim} = 15$N

$\therefore$ $a_{sys} = \frac{5}{6}$ m/s²



$f_1 = 10 \times \frac{5}{6} = \frac{25}{3}$ m/s²

$\therefore$ Case I failed

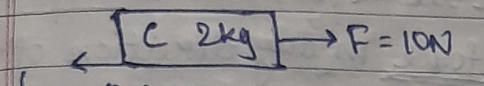
All blocks will not move together.

$\therefore$ $f_1 \leq f_{1\,lim}$

$\frac{25}{3} < 4 \Rightarrow$ False

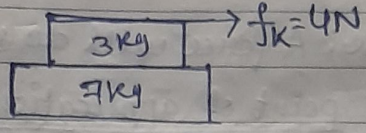
Page No.
Date:

case: Blocks are moving separately:
∴ f_k will act.

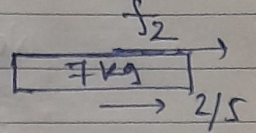


$f_k = 0.2 \times 20$
$= 4N$

$a_c = \frac{10-4}{2} = 3 m/s^2$



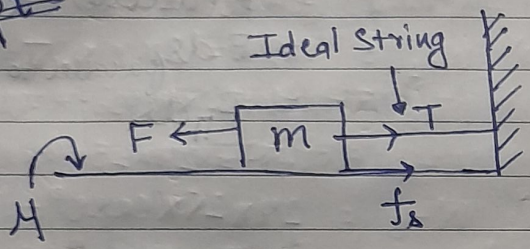
$a_B = \frac{4}{10} = \frac{2}{5} m/s^2$



$f_2 = 7 \times \frac{2}{5} = 2.8N$

$a_A = \frac{2}{5} m/s^2$

concept

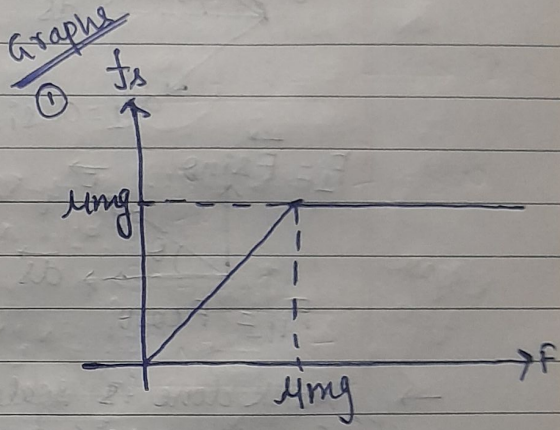


$F = T + f_s$

initially F↑ f_s↑ T=0

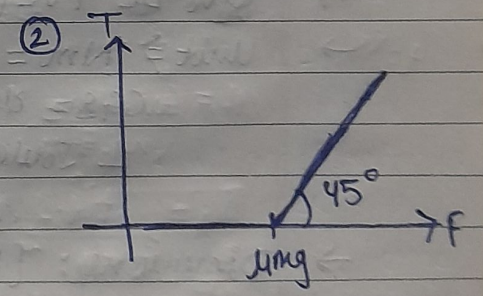
when $f_s = f_{smax}$ ⇒ F↑ f_{smax} T↑

Graphs

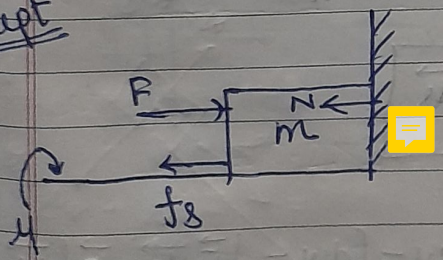


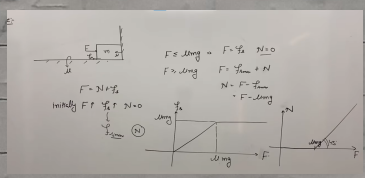
∴

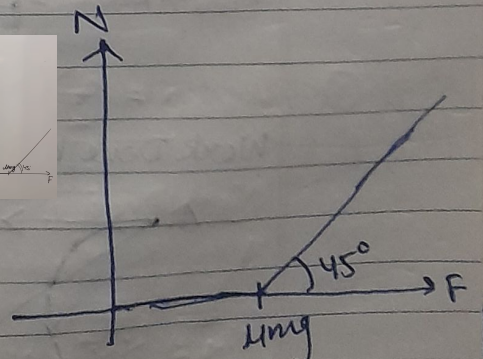
$F \le \mu mg$ ⇒	$F = f_s$	$T = 0$
$F > \mu mg$ ⇒	$F = f_{smax} + T$	
	$T = F - f_{smax}$	
	$T = F - \mu mg$	



concept







$F \le \mu mg$ ⇒	$F = f_s$ $N = 0$
$F > \mu mg$ ⇒	$F = f_{smax} + N$
	$N = F - f_{smax}$
	$N = F - \mu mg$